NON-MF GROUPS

SAUERS

Some groups are not MF. We construct groups for which every homomorphism to an MF group is trivial. The obstruction is a property-(T) subgroup conjugated into a proper subgroup of itself, together with an element that commutes with the subgroup while its conjugate does not: such elements have asymptotic matrix images that vanish in Hilbert–Schmidt norm, and property (T) of a normal subgroup they generate upgrades that to vanishing in operator norm. If a countable unital ring R contains s, t with $ts = 1$ and $R(1 - st)R = R$, then every homomorphism from $\mathrm{EL}_n(R)$ to an MF group is trivial for all $n \geq 2$. In particular, the unit group H of the binary Leavitt algebra $L_{\mathbb{F}_2}(1, 2)$ is a finitely generated simple property-(T) group that is not MF, and $C^*_{\mathrm{r}}(H)$ is separable, stably finite, and not MF. For a countable purely infinite simple ring the MF radical of $\mathrm{GL}_n(R)$ is its commutator subgroup and the largest MF quotient is $K_1(R)$. A finite-block construction gives a finitely presented, centerless, sofic group E whose MF radical is exactly its perfect, residually finite block subgroup $*_{j\geq 1}(A_5)^8$, with residually finite quotient; it splits as $J \rtimes \mathbb{Z}$ with J locally residually finite, so its canonical trace on $C^*_{\max}(E)$ is amenable but not quasidiagonal. Its proof extends Kazhdan transport to fixed points in finite-dimensional invariant C^*-subalgebras. A single cyclic HNN extension then gives a finitely presented sofic group whose MF quotients are exactly its cyclic quotients, while its reduced C^*-algebra is simple with a unique trace, has stable rank one, and is not MF. One amalgamation of that group realizes any finitely presented sofic group as the largest MF quotient of a finitely presented sofic non-MF group, preserving its profinite completion and its rational cohomology. Finally, we construct a two-generated, finitely presented, torsion-free, acylindrically hyperbolic property-(T) group with no nontrivial homomorphism to an MF group.

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1. Introduction

We work with countable groups throughout. A countable group G is *MF* [CDE] if there are positive integers d_n and maps $V_n\colon G \to \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \qquad (g \in G \setminus \{1\}).$$

In other words, G is MF if there are finite unitary matrices, one for each element of G, that multiply correctly up to an error tending to zero in operator norm while

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the matrices assigned to a nonidentity element do not converge to the identity. For countable G, the second condition may equivalently be required along the full sequence with a constant independent of g [Kor, Propositions 2 and 7]. The strong convergence convention of [GKM, Sch] also requires the models to reproduce the operator norms of the left regular representation, so a group that is not MF as defined here is not MF in that convention either. Equivalently, G embeds in the unitary group of the norm matrix corona

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here and below, products of C^*-algebras mean bounded products, and $\bigoplus$ denotes the ideal of norm-null sequences. The quotient $\mathcal{Q}_{\mathbf{d}}$ is the corona algebra of $\bigoplus_n M_{d_n}(\mathbb{C})$, since $\prod_n M_{d_n}(\mathbb{C})$ is its multiplier algebra. We call a homomorphism $G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ a *corona homomorphism*. A separable C^*-algebra is MF if it embeds in a norm matrix corona [BK]. Write

$$\operatorname{Rad}_{\mathrm{MF}}(G) = \bigcap \{\ker f : f\colon G \to M,\ M \text{ an MF group}\}$$

for the *MF radical* of G. For countable G the quotient $G/\operatorname{Rad}_{\mathrm{MF}}(G)$ is again MF, because countably many homomorphisms to MF groups separate the elements outside the radical and the direct sums of their models along a common sequence of matrix sizes remain asymptotically multiplicative; so it is the largest MF quotient of G.

Some groups admit no such approximation. We construct groups for which every homomorphism to an MF group is trivial, as well as a sofic group with a nontrivial element killed by every MF homomorphism. The proof uses a property-(T) subgroup conjugated into a proper subgroup of itself. For finitely generated groups, a separate tensor argument shows that each element killed by all MF homomorphisms has a finite linear certificate in operator norm (Theorem 2.10).

Our first counterexample is the group

$$W = \mathrm{Cl}(X) \rtimes V,$$

where V is the ascending HNN extension of $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ along $(v, A) \mapsto (2v, A)$, the set X is the coset space of $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ in V, and $\mathrm{Cl}(X)$ is generated by involutions c_x, $x \in X$, and a central involution ε, with $[c_x, c_y] = \varepsilon$ for $x \neq y$. Every homomorphism from W to an MF group kills ε (Section 4).[1] This paper focuses on a different counterexample,

$$H = L_{\mathbb{F}_2}(1,2)^{\times},$$

the unit group of the binary Leavitt algebra, because H is simple and every homomorphism from H to an MF group is trivial. The right module R_R over $R = L_{\mathbb{F}_2}(1,2)$ satisfies $R_R \cong R_R \oplus R_R$, and H is its automorphism group; Khanh–Thanh show that $H \cong \mathrm{GL}_n(R) = \mathrm{EL}_n(R)$ for every $n \geq 2$ [K–T, Proposition 4.2 and Corollary 4.4]. OpenAI used this group to construct the first nonsofic group [OAI, Chapter 3].

We use the group commutator convention $[g,h] = ghg^{-1}h^{-1}$. The groups here fail to be MF because they contain a property-(T) subgroup that is conjugated into a

[1] The Lean proof was registered at Palomar on August 24, 2026: https://palomar-registry.org/entry?id=PALOMAR-2026-08-24-000006&version=1.

proper subgroup of itself, together with an element that commutes with the subgroup while its conjugate does not. For $L \leq G$ put

$$\mathrm{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}.$$

The elements of $\mathrm{Comp}_G(L)$ are the compressors of L. An element c of the centralizer $C_G(L)$ commutes with L, so ucu^{-1} commutes with uLu^{-1} but need not commute with the rest of L. The normal subgroup of G generated by the commutators of all such elements ucu^{-1} with elements of L is

$$\mathfrak{D}_G(L) = \langle\!\langle [ucu^{-1}, \ell] : u \in \mathrm{Comp}_G(L),\ c \in C_G(L),\ \ell \in L \rangle\!\rangle_G. \tag{1}$$

For every homomorphism $f : G \to H$,

$$f(\mathfrak{D}_G(L)) \leq \mathfrak{D}_H(f(L)), \tag{2}$$

since compressors, centralizing elements, and commutators of L map to compressors, centralizing elements, and commutators of $f(L)$, and conjugates map to conjugates.

Theorem A (one-sided compression criterion)**.** *Let G be countable and let $L \leq G$ have property* (T)*. If $K \trianglelefteq G$ has property* (T) *and $K \leq \mathfrak{D}_G(L)$, then every homomorphism from G to an MF group is trivial on K. In particular, if such a K is nontrivial, then G is not MF, and if G has property* (T) *and $\mathfrak{D}_G(L) = G$, then every homomorphism from G to an MF group is trivial.*

Asymptotic multiplicativity in operator norm makes the conjugation maps $\mathrm{Ad}(V_n(g)) : x \mapsto V_n(g)xV_n(g)^*$ an asymptotically multiplicative family of unitaries on $M_{d_n}(\mathbb{C})$ with its normalized Hilbert–Schmidt inner product. Here $\|\mathrm{Ad}(A) - \mathrm{Ad}(B)\| \leq 2\|A - B\|$, with the norm on the left taken in the operator algebra of this Hilbert space. The classes of these maps form a homomorphism of G into the unitary group of the norm matrix corona with coordinate sizes d_n^2. Let U be the class of $(\mathrm{Ad}(V_n(u)))$ and P the image of the Kazhdan projection of L in that corona. Then U^*PU projects onto the vectors fixed by $u^{-1}Lu$, a subgroup containing L because $uLu^{-1} \leq L$; so $U^*PU \leq P$, and stable finiteness of that corona forces $U^*PU = P$. So conjugation by u preserves the Hilbert–Schmidt asymptotic commutant of L, and the matrices assigned to an element of $\mathfrak{D}_G(L)$ tend to 1 in Hilbert–Schmidt norm in every operator norm asymptotic representation (Corollary 2.5). Property (T) of K turns this into triviality in operator norm. A corona homomorphism nontrivial on K compresses to a corner where the Kazhdan projection of K vanishes. A limit of the normalized traces of that corner is a tracial state taking the value 1 on every element of K, hence is the trivial character; but the trivial character takes the value 1 on the Kazhdan projection, which is 0 there (Theorem 2.7).[2]

The stable letter of an ascending HNN extension of a property-(T) group along a proper self-embedding is a compressor. For a unital ring R, write $e_{ij}(a) = 1 + aE_{ij}$ and let $\mathrm{EL}_n(R)$ be the subgroup of $\mathrm{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$. A pair $s, t \in R$ with $ts = 1$ gives a compressor in the same way: an explicit element $u \in \mathrm{EL}_4(R)$ conjugates $\mathrm{EL}_3(R)$ into itself by $e_{ij}(a) \mapsto e_{ij}(sat)$, and if $1 - st$ generates R as a

[2] A Hilbert–Schmidt bound on the multiplicative defect does not give an operator norm bound on the conjugation maps, so this argument does not apply to sofic or hyperlinear approximations.

two-sided ideal, then one commutator $[ucu^{-1}, \ell]$ normally generates all of $\mathrm{EL}_4(R)$. This happens once, in the universal unital associative ring $\mathcal{C} = \mathbb{Z}\langle s_0, s_1, t_0, t_1\rangle/(t_i s_j - \delta_{ij} : i, j \in \{0, 1\})$, and $\mathrm{EL}_4(\mathcal{C})$ then maps to every $\mathrm{EL}_n(R)$ below with normally generating image.

Theorem B (full complementary idempotents)**.** *Let R be a countable unital associative ring. Suppose that $s, t \in R$ satisfy*

$$ts = 1, \qquad R(1 - st)R = R.$$

For every $n \geq 2$, every homomorphism from $\mathrm{EL}_n(R)$ to an MF group is trivial. Every homomorphism from $\mathrm{EL}_4(\mathcal{C})$ to an MF group is trivial; that group is finitely generated and has property (T)*; and there is a homomorphism from it to $\mathrm{EL}_n(R)$ whose image normally generates $\mathrm{EL}_n(R)$.*

Here $R(1 - st)R$ is the two-sided ideal generated by $1 - st$, so the hypothesis says that $\sum_j a_j(1 - st)b_j = 1$ for finitely many $a_j, b_j \in R$; we call such an idempotent $1 - st$ *full*. Equivalently, R contains two disjoint copies of itself: elements v_0, v_1, w_0, w_1 with $w_i v_j = \delta_{ij}$ (Lemma 3.1). A unital ring is *directly finite* if $ts = 1$ implies $st = 1$. The hypothesis holds in every countable simple unital ring that is not directly finite and in every Leavitt algebra $L_k(1, m)$ over a countable field k, $m \geq 2$ (Corollary 3.4). The ring R need not be finitely generated. In the following example $\mathrm{EL}_n(R)$ is moreover simple. Let $L_{\mathbb{F}_2}(1, 2)$ denote the binary Leavitt algebra, the unital $\mathbb{F}_2$-algebra generated by s_0, s_1, t_0, t_1 subject to

$$t_i s_j = \delta_{ij}, \qquad s_0 t_0 + s_1 t_1 = 1. \tag{3}$$

Theorem C (the unit group of the binary Leavitt algebra)**.** *Let $R = L_{\mathbb{F}_2}(1, 2)$ and let $H = R^\times$ be its unit group. Then $H \cong \mathrm{EL}_4(R)$, and H is finitely generated, nontrivial, simple, has property* (T)*, and every homomorphism from H to an MF group is trivial. So H is not MF, and $C^*_{\mathrm{r}}(H)$ is separable and stably finite but not MF, while $C^*_{\max}(H)$ is not finite: it contains a proper isometry.*

For a countable purely infinite simple ring R and $n \geq 1$, Theorem 3.6 identifies the intersection of the kernels of all homomorphisms from $\mathrm{GL}_n(R)$ to MF groups with the commutator subgroup, and the quotient with $K_1(R)$, which is MF. For the unit group of $L_k(1, d)$ the quotient is $k^\times/(k^\times)^{d-1}$.

Theorem D (a torsion-free finitely presented example)**.** *There is a two-generated, finitely presented, torsion-free, acylindrically hyperbolic group Q with property* (T) *such that every homomorphism from Q to an MF group is trivial. In particular, no nontrivial quotient of Q is MF.*

Section 7 builds Q from Fournier-Facio's torsion-free property-(T) group [FF, §2] and Hull's small cancellation theorem [Hu].

Theorem E (a finitely presented sofic group with exact MF radical)**.** *There is a finitely presented, centerless, sofic group E in a split exact sequence*

$$1 \longrightarrow B \longrightarrow E \longrightarrow V \longrightarrow 1, \qquad B \cong \mathop{*}_{j \geq 1} (A_5)^8,$$

where V is residually finite and

$$\mathrm{Rad}_{\mathrm{MF}}(E) = \bigcap_{\substack{N \trianglelefteq E \\ [E:N] < \infty}} N = B \neq 1.$$

The subgroup B is perfect and residually finite, and any nonidentity element of one coordinate A_5 normally generates it in E. Moreover, $E = J \rtimes \mathbb{Z}$ with J locally residually finite. The canonical trace on $C^*_{\max}(J)$ is quasidiagonal, while the canonical trace on $C^*_{\max}(E)$ is amenable and not quasidiagonal.

Section 5 gives a finite presentation and computes the radical. Thus both the kernel B and the quotient V are residually finite, although E is not MF. The construction uses commuting lamps within blocks of eight sites. Keeping different blocks in a free product makes finite presentability possible. The analytic step is Theorem 5.1: a finite-dimensional invariant algebra has the same fixed points under a Kazhdan subgroup and its conjugate compressed copy.

The trace on $C^*_{\max}(E)$ answers the question of Brown [Br, discussion preceding Proposition 3.5.1] and of Schafhauser, Tikuisis and White [STW, Problem X(1)] whether every amenable trace is quasidiagonal. Tikuisis, White and Winter proved that faithful traces on separable nuclear C^*-algebras satisfying the universal coefficient theorem are quasidiagonal [TWW]; here $C^*_{\max}(E)$ is not nuclear, since E contains a Kazhdan subgroup and is not amenable. The canonical trace of $C^*_{\max}(J)$ is quasidiagonal, so the crossed product by $\mathbb{Z}$ destroys quasidiagonality of the canonical trace. Since J is a direct limit of residually finite groups it is MF [Kor, Corollary 10 and Proposition 13], so MF groups are not closed under semidirect products with $\mathbb{Z}$, which answers a question of Korchagin [Kor, remark following Proposition 12].

The finite-block example also permits an exact calculation after one HNN extension. In the next theorem, the radical has no nontrivial MF images even when considered as a group in its own right.

Theorem F (a sofic group with exactly cyclic MF quotients)**.** *There is a finitely presented, sofic, acylindrically hyperbolic group P with a split epimorphism $\chi\colon P \to \mathbb{Z}$ and an explicit infinite-order element d such that*

$$N := \ker\chi = \langle\langle d \rangle\rangle_P = \mathrm{Rad}_{\mathrm{MF}}(P) = \bigcap_{\substack{R \trianglelefteq P \\ [P:R] < \infty}} R, \qquad \mathrm{Rad}_{\mathrm{MF}}(N) = N \neq 1.$$

The kernel N is perfect and is not finitely generated. For every normal subgroup $R \trianglelefteq P$,

$$\mathrm{Rad}_{\mathrm{MF}}(P/R) = NR/R, \qquad P/R \text{ is MF} \iff d \in R \iff P/R \text{ is cyclic.}$$

Every homomorphism from P to an MF group factors uniquely through χ, and $\widehat{P} \cong \widehat{\mathbb{Z}}$. The group P is a Powers group; in particular, $C^*_{\mathrm{r}}(P)$ is simple, has a unique tracial state, has stable rank one, and is not MF.

Theorem 6.3 makes P a device: amalgamating it with any finitely presented sofic group A along an infinite-order element produces a finitely presented sofic non-MF group whose largest MF quotient is A, with the profinite completion and the rational cohomology of A unchanged.

Section 6 constructs P by adding one generator and one relator to the presentation of E. The relation identifies a normal generator of the quotient V with an infinite-order word in B. The two examples differ intrinsically:

$$\mathrm{Rad}_{\mathrm{MF}}(E) = B, \quad \mathrm{Rad}_{\mathrm{MF}}(B) = 1, \qquad \mathrm{Rad}_{\mathrm{MF}}(P) = N, \quad \mathrm{Rad}_{\mathrm{MF}}(N) = N,$$

since B is residually finite while N has no nontrivial MF image at all. The added relation forces all MF maps to kill E, while soficity is preserved by the cyclic HNN extension theorem of Collins and Dykema [CD, Corollary 3.6].

The examples answer different questions, and the following table records what each one contributes.

Object	Homomorphisms to MF groups	Contribution
$H = L_{\mathbb{F}_2}(1,2)^\times$	all trivial (Theorem C)	simple, Kazhdan, and a module automorphism group
$\mathrm{GL}_n(R)$, R purely infinite simple	factor through $K_1(R)$ (Theorem 3.6)	a classification, not just an example
W	all kill ε (Proposition 4.3)	lamps over a proper self-embedding
E	factor through V, radical B (Theorem E)	finitely presented, with radical and quotient both residually finite, and an amenable trace that is not quasidiagonal
P	factor through $\mathbb{Z}$, radical N (Theorem F)	the radical has no nontrivial MF image of its own
Q	all trivial (Theorem D)	two-generated, finitely presented, torsion-free
G_A	factor through A (Theorem 6.3)	installs a prescribed largest MF quotient on a chosen A

Related work. Blackadar and Kirchberg introduced the notions of MF and NF algebras in a 1994 Oberwolfach abstract [BK94]. They wrote that the classes of NF, strong NF, and separable nuclear stably finite C^*-algebras might coincide. Their 1997 paper developed these notions and proved that a separable C^*-algebra is NF if and only if it is nuclear and MF [BK]. It also asked whether every separable nuclear stably finite C^*-algebra is quasidiagonal; for algebras that satisfy the universal coefficient theorem and have a faithful trace, the theorem of Tikuisis, White, and Winter recalled above answers this [TWW]. The question without nuclearity, whether every stably finite C^*-algebra is MF, is the MF problem, also posed by Blackadar and Kirchberg [GH, §6]. A positive answer would make every countable group MF, because the reduced group C^*-algebra is separable and stably finite; Lubotzky and Oppenheim and Thom noted this when they listed operator norm approximation of groups among the open approximation problems [LO, Th]. Carrión, Dadarlat, and Eckhardt introduced MF groups [CDE], and whether every countable group is MF remained open [Kor, BDL]. The MF problem itself was answered negatively in 2020: the failure of the Connes embedding problem [JNVWY] yields a II_1 factor that embeds in no norm ultraproduct of matrix algebras, hence a separable stably finite C^*-algebra that is not MF [GH, Proposition 6.1 and Remark 6.2]. Theorem C gives a counterexample among reduced group C^*-algebras and answers the group question.

Theorem A uses the configuration of the nonsofic construction [OAI, Proposition 2.3]: a property-(T) subgroup L conjugated into itself by an element u, and an element c commuting with L whose conjugate ucu^{-1} does not. There, soficity and the rigidity theorems of Kun [Ku16] and Kun–Thom [KT19] force the centralizing subgroup to be locally embeddable into finite groups, and a copy of Thompson's group V gives the contradiction. Here the rigidity comes from the Kazhdan projection of L in a norm matrix corona, which commutes with the compressor because the corona is stably finite, and the contradiction is a commutator $[ucu^{-1}, \ell] \neq 1$ that every operator norm asymptotic representation sends to 1 in Hilbert–Schmidt norm. The group H is the group of [OAI], and Theorem D reuses Fournier-Facio's configuration [FF]. Bachner–Dogon–Lubotzky showed that for groups of Deligne type, operator–Hilbert–Schmidt stability would imply that the group is not MF [BDL, Proposition 1.5]. Here the compression relation replaces the stability hypothesis.

Leavitt introduced the algebras $L_k(1,n)$ [Lev]. The binary Leavitt algebra is purely infinite simple [AA], is an exchange ring [Ara], and has center $\mathbb{F}_2$ [AC]. Preusser's sandwich theorem then gives simplicity of H [Pre], and property (T) is Ershov–Jaikin-Zapirain's theorem [EJZ, Theorem 1.1].

2. One-sided compression

2.1. **Corona homomorphisms.**

Lemma 2.1. *Let G be countable and $K \leq G$. Every homomorphism from G to an MF group is trivial on K if and only if every corona homomorphism from G is trivial on K. Let A be a unital C^*-algebra and let $\pi\colon G \to \mathcal{U}(A)$ be an injective group homomorphism. If A embeds in a norm matrix corona, then G is MF. In particular, this applies to the canonical group homomorphisms into $\mathcal{U}(C^*_{\max}(G))$ and $\mathcal{U}(C^*_{\mathrm{r}}(G))$.*

Proof. The image of a corona homomorphism is a countable MF group. Conversely, compose a homomorphism to an MF group with a corona embedding of its image; the resulting corona homomorphism has the same kernel. If ι is an embedding as in the last assertion, then $g \mapsto \iota(\pi(g)) + 1 - \iota(1)$ is a corona homomorphism that is injective on G; the complement term allows ι to be nonunital. □

2.2. **Kazhdan transport in normalized Hilbert–Schmidt norm.**

For an actual representation $\rho\colon G \to \mathcal{U}(d)$ and $u \in G$ with $uLu^{-1} \leq L$, the commutant $\mathcal{C} = \rho(L)' \subseteq M_d(\mathbb{C})$ satisfies $\rho(u)^*\mathcal{C}\rho(u) \subseteq \mathcal{C}$, and the two spaces have the same finite dimension, so the inclusion is an equality; if c commutes with L, then $\rho(ucu^{-1})$ commutes with $\rho(L)$, and ρ is trivial on $\mathfrak{D}_G(L)$. An asymptotic representation has no exact commutant to count. Property (T) supplies a Kazhdan projection in its place, and stable finiteness of the corona replaces the dimension count.

For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \mathrm{tr}_d(a^*a)^{1/2}, \qquad \mathrm{tr}_d = \frac{1}{d}\,\mathrm{Tr}\,.$$

An *operator norm asymptotic representation* of a group G is a sequence of maps $V_n\colon G \to \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G).$$

The elements g with $\|V_n(g) - 1\|_2 \to 0$ form a normal subgroup of G, because $\|a\|_2 \leq \|a\|$, the Hilbert–Schmidt norm is invariant under adjoints and unitary conjugation, and $\|V_n(g^{-1}) - V_n(g)^*\| \to 0$:

$$\|V_n(gh) - 1\|_2 \leq \|V_n(g) - 1\|_2 + \|V_n(h) - 1\|_2 + o(1),$$

$$\left\|V_n(g^{-1}) - 1\right\|_2 = \|V_n(g) - 1\|_2 + o(1), \qquad \left\|V_n(ghg^{-1}) - 1\right\|_2 = \|V_n(h) - 1\|_2 + o(1).$$

For $L \leq G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \;\middle|\; \begin{array}{l} \sup_n \|x_n\|_2 < \infty, \\ \|V_n(\ell)x_n - x_nV_n(\ell)\|_2 \longrightarrow 0 \quad (\ell \in L) \end{array} \right\}$$

be the Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\mathrm{Ad}(V(g))$, of the Hilbert–Schmidt bounded matrix sequences. A unital C^*-algebra is finite if each of its isometries is unitary, and stably finite if all matrix algebras over it are finite.

Lemma 2.2. *For every sequence of positive integers (m_n), the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Write A for the displayed corona. If $v^*v = 1$ in A and (x_n) is a bounded lift of v, then $\|x_n^*x_n - 1\| \to 0$, so for all large n the matrix $u_n = x_n(x_n^*x_n)^{-1/2}$ is unitary and $\|u_n - x_n\| \to 0$; after assigning arbitrary unitary values to the finitely many remaining coordinates, (u_n) is a unitary lift of v, and $vv^* = 1$. So A is finite, and so is $M_k(A)$, which is the corona with matrix sizes km_n. Finally, if $p \leq q$ are equivalent projections in a finite algebra B and w is a partial isometry with $w^*w = q$ and $ww^* = p$, then $w = qwq$, so $w + (1 - q)$ is an isometry of B and hence unitary; therefore $p + (1 - q) = (w + (1 - q))(w + (1 - q))^* = 1$ and $p = q$. □

For a group L with property (T), the Kazhdan projection is the central projection $e_L \in C^*_{\max}(L)$ whose image in every unitary representation of L is the orthogonal projection onto the L-invariant vectors [AW].

Lemma 2.3 (one-sided order for the Kazhdan projection)**.** *Let L have property* (T)*, let B be a unital C^*-algebra, and let $\pi\colon L \to \mathcal{U}(B)$ be a homomorphism. Let $P \in B$ be the image of e_L under the $*$-homomorphism $C^*_{\max}(L) \to B$ induced by π. If a unitary $U \in B$ satisfies $U\pi(L)U^* \subseteq \pi(L)$, then $U^*PU \leq P$.*

Proof. Fix a faithful nondegenerate representation of B on a Hilbert space $\mathcal{H}$, and let $\operatorname{Fix}\pi(L)$ be the space of vectors fixed by $\pi(L)$. Then P is the orthogonal projection onto $\operatorname{Fix}\pi(L)$. If $\xi \in \operatorname{Fix}\pi(L)$ and $\ell \in L$, then $\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi$, because $U\pi(\ell)U^*$ belongs to $\pi(L)$. So $U^* \operatorname{Fix} \pi(L) \subseteq \operatorname{Fix} \pi(L)$. Since U^*PU is the orthogonal projection onto $U^* \operatorname{Fix} \pi(L)$, it follows that $U^*PU \leq P$. □

Theorem 2.4 (one-sided Kazhdan transport)**.** *Let $L \leq G$ have property* (T)*, and let $u \in G$ satisfy $uLu^{-1} \leq L$. Let (V_n) be an operator norm asymptotic representation of G. Then*

$$\operatorname{Ad}(V(u))(\mathcal{C}_2(V, L)) = \mathcal{C}_2(V, L).$$

Proof. For unitaries A, B,

$$\|\operatorname{Ad}(A) - \operatorname{Ad}(B)\| \leq 2\,\|A - B\|\,,$$

where the norm on the left is the operator norm on the Hilbert space $(M_{d_n}(\mathbb{C}), \|\cdot\|_2)$, so the maps $\operatorname{Ad}(V_n(g))$ are asymptotically multiplicative in operator norm and

$$\widetilde{\sigma}(g) = [\operatorname{Ad}(V_n(g))]_n \in \mathcal{U}(\mathcal{B}), \qquad \mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C}))/\bigoplus_n B(M_{d_n}(\mathbb{C})),$$

is a homomorphism, where $B(M_{d_n}(\mathbb{C}))$ denotes the operators on the Hilbert–Schmidt Hilbert space. The algebra $\mathcal{B}$ is a norm matrix corona with coordinate sizes d_n^2 after a choice of matrix units. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection of L [AW] under $C^*_{\max}(L) \to \mathcal{B}$, and lift P to orthogonal projections P_n on $(M_{d_n}(\mathbb{C}), \|\cdot\|_2)$ by functional calculus.

A Hilbert–Schmidt bounded sequence (x_n) lies in $\mathcal{C}_2(V, L)$ if and only if

$$\|P_n x_n - x_n\|_2 \longrightarrow 0.$$

In one direction, fix $\varepsilon > 0$ and, by density of the group algebra in $C^*_{\max}(L)$, a finite set $F \subseteq L$ and scalars $(a_\ell)_{\ell \in F}$ with

$$\Big\|\sum_{\ell \in F} a_\ell u_\ell - e_L\Big\| < \varepsilon,$$

where u_ℓ is the canonical unitary of ℓ; the trivial character, which sends e_L to 1, gives $\big|\sum_{\ell \in F} a_\ell - 1\big| < \varepsilon$. The element $\sum_{\ell \in F} a_\ell \widetilde{\sigma}(\ell)$ of $\mathcal{B}$ is within ε of P, so

$$\limsup_n \left\| P_n - \sum_{\ell \in F} a_\ell \operatorname{Ad}(V_n(\ell)) \right\| \leq \varepsilon.$$

If $(x_n) \in \mathcal{C}_2(V, L)$ and $\sup_n \|x_n\|_2 \leq c$, then $\operatorname{Ad}(V_n(\ell))x_n = x_n + o(1)$ in $\|\cdot\|_2$ for each $\ell \in F$, so

$$\limsup_n \|P_n x_n - x_n\|_2 \leq \varepsilon c + \Big|\sum_{\ell \in F} a_\ell - 1\Big| c \leq 2\varepsilon c,$$

and ε was arbitrary. In the other direction, $u_\ell e_L = e_L$ in $C^*_{\max}(L)$, so $\widetilde{\sigma}(\ell)P = P$ and $\|\operatorname{Ad}(V_n(\ell))P_n - P_n\| \to 0$; if $\|P_n x_n - x_n\|_2 \to 0$, then

$$\begin{aligned}\|\operatorname{Ad}(V_n(\ell))x_n - x_n\|_2 &\leq 2\,\|x_n - P_n x_n\|_2 \\ &\quad + \|\operatorname{Ad}(V_n(\ell))P_n - P_n\|\;\|x_n\|_2 \longrightarrow 0.\end{aligned}$$

Put $U = \widetilde{\sigma}(u)$. Since $uLu^{-1} \leq L$, $U\widetilde{\sigma}(L)U^* \subseteq \widetilde{\sigma}(L)$, so $U^*PU \leq P$ by Lemma 2.3, and $U^*PU = P$ by Lemma 2.2, since the two projections are unitarily equivalent. So $[U, P] = 0$ and $\|[\mathrm{Ad}(V_n(u)), P_n]\| \to 0$. For $(x_n) \in \mathcal{C}_2(V, L)$,

$$\left\| P_n \,\mathrm{Ad}(V_n(u))^{\pm 1} x_n - \mathrm{Ad}(V_n(u))^{\pm 1} x_n \right\|_2 \leq \left\| [P_n, \mathrm{Ad}(V_n(u))^{\pm 1}] \right\| \, \|x_n\|_2 + \|P_n x_n - x_n\|_2 \longrightarrow 0,$$

so $\mathrm{Ad}(V(u))^{\pm 1} x \in \mathcal{C}_2(V, L)$ by the displayed equivalence. $\square$

Corollary 2.5. *Let $L \leq G$ have property* (T) *and let (V_n) be an operator norm asymptotic representation of G. Then*

$$\|V_n(d) - 1\|_2 \longrightarrow 0 \qquad (d \in \mathfrak{D}_G(L)).$$

Proof. Let $u \in \mathrm{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, and write $\mathcal{C}_2 = \mathcal{C}_2(V, L)$. Since c commutes with L, $(V_n(c)) \in \mathcal{C}_2$, so $(V_n(u)V_n(c)V_n(u)^*) \in \mathcal{C}_2$ by Theorem 2.4, and then $(V_n(ucu^{-1})) \in \mathcal{C}_2$ because $\|V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1})\| \to 0$. So $\|V_n(\ell)V_n(ucu^{-1}) - V_n(ucu^{-1})V_n(\ell)\|_2 \to 0$, and by asymptotic multiplicativity $\|V_n([ucu^{-1}, \ell]) - 1\|_2 \to 0$. The elements with this property form a normal subgroup of G, so it contains $\mathfrak{D}_G(L)$. $\square$

2.3. **From Hilbert–Schmidt to operator norm.** Smallness in normalized Hilbert–Schmidt norm does not bound the operator norm: for

$$D_d = \mathrm{diag}(-1, 1, \ldots, 1) \in \mathcal{U}(d), \qquad \|D_d - 1\|_2 = \frac{2}{\sqrt{d}} \longrightarrow 0, \qquad \|D_d - 1\| = 2.$$

A corner occupying a vanishing fraction of the matrix can therefore carry the entire obstruction, which is why the corner below is renormalized by its own rank.

Restricting a corona homomorphism ρ to a corner requires a correction: a projection q commuting with $\rho(G)$ has lifts q_n that commute only asymptotically with unitary lifts $U_n(g)$ of $\rho(g)$, so the compressions $q_n U_n(g) q_n$ are only approximately unitary in the matrix corners.

Lemma 2.6 (central corona corners)**.** *Let $\rho\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. After passing to an infinite subsequence of coordinates, there are ranks $r_n \geq 1$, an identification of the corner $q\mathcal{Q}_{\mathbf{d}}q$ with the norm matrix corona $\mathcal{Q}_{\mathbf{r}}$, and an operator norm asymptotic representation $W_n\colon G \to \mathcal{U}(r_n)$ whose corona homomorphism is $g \mapsto q\rho(g)$ under this identification.*

Proof. Lift q to projections $q_n \in M_{d_n}(\mathbb{C})$ by functional calculus. Since $q \neq 0$, infinitely many q_n are nonzero; retain those coordinates and put $r_n = \mathrm{rank}(q_n)$. Identifying each corner $q_n M_{d_n}(\mathbb{C}) q_n$ with $M_{r_n}(\mathbb{C})$ identifies the corner $q\mathcal{Q}_{\mathbf{d}}q$ with the corona $\mathcal{Q}_{\mathbf{r}}$: a bounded sequence (z_n) representing $z \in \mathcal{Q}_{\mathbf{d}}$ gives the representative $(q_n z_n q_n)$ of qzq. Since q commutes with $\rho(G)$, the map $g \mapsto q\rho(g)$ is a homomorphism from G to the unitary group of the corner, with unit q. Each value is a unitary of $\mathcal{Q}_{\mathbf{r}}$, so, as in the proof of Lemma 2.2, it lifts to unitaries $W_n(g) \in \mathcal{U}(r_n)$, with $W_n(1) = I_{r_n}$. Multiplicativity in the corona makes (W_n) an operator norm asymptotic representation whose corona homomorphism is $g \mapsto q\rho(g)$. $\square$

Theorem 2.7 (normal Kazhdan subgroups)**.** *Let G be countable and let $K \trianglelefteq G$ have property* (T)*. If every operator norm asymptotic representation* (V_n) *of G satisfies* $\|V_n(k) - 1\|_2 \to 0$ *for all $k \in K$, then every corona homomorphism from G is trivial on K.*

Proof. Suppose that a corona homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K. Let $e_K \in C^*_{\max}(K)$ be the Kazhdan projection of K, and let $p \in \mathcal{Q}_{\mathbf{d}}$ be its image under the homomorphism $C^*_{\max}(K) \to \mathcal{Q}_{\mathbf{d}}$ induced by $\Theta|_K$. In any faithful representation of the corona, p is the projection onto the K-fixed vectors. Since $gKg^{-1} = K$, the range of $\Theta(g)p\Theta(g)^*$ is $\Theta(g)\operatorname{Fix}\Theta(K) = \operatorname{Fix}\Theta(gKg^{-1}) = \operatorname{Fix}\Theta(K)$, so

$$\Theta(g)p\Theta(g)^* = p \qquad (g \in G),$$

and $q = 1-p$ is nonzero because Θ is nontrivial on K. By Lemma 2.6 there is an operator norm asymptotic representation $W_n\colon G \to \mathcal{U}(r_n)$ whose corona homomorphism $\widehat{\Theta}$ is the coordinate restriction of $g \mapsto q\Theta(g)$. Let $\pi\colon C^*_{\max}(K) \to \mathcal{Q}_{\mathbf{r}}$ be the homomorphism induced by $\widehat{\Theta}|_K$. Since q commutes with $\Theta(K)$, the map $a \mapsto q\Theta(a)$ is a homomorphism on $C^*_{\max}(K)$ agreeing with π on K, so $\pi(e_K)$ is the coordinate restriction of $qp = 0$.

Fix a free ultrafilter ω and let τ be the limit along ω of the normalized traces of the coordinates $M_{r_n}(\mathbb{C})$. A norm-null sequence has vanishing traces, since $|\operatorname{tr}_{r_n}(x)| \le \|x\|$, so τ is a well defined tracial state on $\mathcal{Q}_{\mathbf{r}}$. The hypothesis applies to (W_n) itself, so

$$|\operatorname{tr}_{r_n}(W_n(k)) - 1| \le \|W_n(k) - I_{r_n}\|_2 \longrightarrow 0 \qquad (k \in K)$$

in the normalized Hilbert–Schmidt norm of $M_{r_n}(\mathbb{C})$, and therefore $\tau(\pi(u_k)) = 1$ for every $k \in K$, where u_k is the canonical unitary of k. The trivial character χ of K is the state of $C^*_{\max}(K)$ with $\chi(u_k) = 1$ for every k, and it has $\chi(e_K) = 1$. The two states $\tau \circ \pi$ and χ agree on every u_k, hence by linearity on the dense subalgebra $\mathbb{C}[K]$, hence on $C^*_{\max}(K)$. So

$$1 = \chi(e_K) = \tau(\pi(e_K)) = \tau(0) = 0,$$

a contradiction. ☐

Proof of Theorem A. By Corollary 2.5, every operator norm asymptotic representation of G satisfies $\|V_n(k) - 1\|_2 \to 0$ for $k \in K \le \mathfrak{D}_G(L)$. By Theorem 2.7, every corona homomorphism is then trivial on K, and by Lemma 2.1 so is every homomorphism to an MF group. If $K \ne 1$, then G does not embed in an MF group, so G is not MF. The last assertion is the case $K = G = \mathfrak{D}_G(L)$. ☐

2.4. **The maximal group C^*-algebra.**

Proposition 2.8 (proper one-sided compression)**.** *Let G be a countable group containing a property-*(T) *subgroup Γ and an element t with $t\Gamma t^{-1} \subsetneq \Gamma$. Then $C^*_{\max}(G)$ contains a proper isometry. In particular, $C^*_{\max}(G)$ is not stably finite, admits no faithful tracial state, and is neither residually finite-dimensional nor MF.*

Proof. Let $P \in C^*_{\max}(G)$ be the image of the Kazhdan projection of Γ [AW] under the canonical map $C^*_{\max}(\Gamma) \to C^*_{\max}(G)$, and let u be the canonical unitary of t. Conjugation by u implements the isomorphism $\Gamma \to t\Gamma t^{-1}$ on canonical unitaries, so by Lemma 2.3 applied with $U = u$, $P \le uPu^*$, where uPu^* is the image of the Kazhdan projection of $t\Gamma t^{-1}$. The inequality is strict: in the quasi-regular representation of G on $\ell^2(G/t\Gamma t^{-1})$, the point mass at the base coset is fixed by $t\Gamma t^{-1}$ and moved by every element of $\Gamma \setminus t\Gamma t^{-1}$, so the two fixed-space projections differ. Put $q = uPu^*$, so that $P < q$, $Pq = qP = P$, $uP = qu$, and $Pu^*(1-q) = Pu^* - Pu^*uPu^* = 0$, and put $s = Pu^* + (1-q)$. Then

$$s^*s = uPu^* + (1-q) = 1, \qquad ss^* = P + (1-q) = 1 - (q-P) \neq 1,$$

so s is a proper isometry, and $\mathrm{diag}(s, 1, \ldots, 1)$ is one in every matrix algebra over $C^*_{\max}(G)$, which is therefore not stably finite. A tracial state τ has $\tau(q-P) = \tau(s^*s) - \tau(ss^*) = 0$ with $q - P$ a nonzero positive element, so no tracial state is faithful. MF algebras are stably finite, and a separable residually finite-dimensional C^*-algebra is MF [BK]. $\square$

A group can satisfy the hypothesis of Proposition 2.8 and be MF. The ascending HNN extension V of $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ along $(v, A) \mapsto (2v, A)$, used in Section 4, has the faithful matrix realization

$$V \cong \left\{ \begin{pmatrix} 2^k A & v \\ 0 & 1 \end{pmatrix} : A \in \mathrm{SL}_3(\mathbb{Z}),\ v \in \mathbb{Z}[1/2]^3,\ k \in \mathbb{Z} \right\},$$

and reduction modulo odd integers separates its elements, so V is residually finite and MF [Kor, Corollary 10]. So the lamps of Section 4 are necessary for the non-MF conclusion there.

2.5. **Finite certificates for individual words.** The obstruction admits a finite, quantitative form even when it kills only part of the group. The finite relation set and threshold below are obtained by a contradiction argument; no explicit list of relators or numerical threshold for the examples is asserted. The following tensor estimate turns a single separation threshold into a linear bound.

Lemma 2.9 (tensor amplification)**.** *If $A \in \mathcal{U}(d)$ and $t = \|A - 1\| > 0$, there is an integer $1 \leq k \leq \pi/t$ with $\left\|A^{\otimes k} - 1\right\| \geq \sqrt{2}$. For any $A, B \in \mathcal{U}(d)$ and $k \geq 1$,*

$$\left\|A^{\otimes k} - B^{\otimes k}\right\| \leq k \|A - B\|.$$

Proof. Since $A - 1$ is normal, some eigenvalue $e^{i\theta}$ of A, with $0 < |\theta| \leq \pi$, has $|e^{i\theta} - 1| = t$. Put $k = \lceil \pi/2|\theta| \rceil$. Then $k|\theta| \geq \pi/2$, and $k|\theta| \leq \pi$ as well: if $|\theta| > \pi/2$ then $k = 1$, and otherwise $k|\theta| < \pi/2 + |\theta| \leq \pi$. So $k\theta/2 \in [\pi/4, \pi/2]$ up to sign and

$$|e^{ik\theta} - 1| = 2|\sin(k\theta/2)| \geq \sqrt{2}.$$

Tensoring an eigenvector for $e^{i\theta}$ shows that $e^{ik\theta}$ is an eigenvalue of $A^{\otimes k}$. Finally $t = 2|\sin(\theta/2)| \leq |\theta|$, so $kt \leq k|\theta| \leq \pi$. For the second assertion, telescope the tensor factors:

$$A^{\otimes k} - B^{\otimes k} = \sum_{j=0}^{k-1} A^{\otimes j} \otimes (A - B) \otimes B^{\otimes (k-1-j)}.$$

Each summand has norm $\|A - B\|$. $\square$

Theorem 2.10 (linear certificates for MF-invisible words)**.** *Let $G = F_m/N$ be a finitely generated group and let $w \in F_m$. Every homomorphism from G to an MF group kills the image of w if and only if there are a finite set $\mathcal{R} \subseteq N$ and $C > 0$ such that, in every positive dimension and for every unitary tuple $U = (U_1, \dots, U_m)$,*

$$\|w(U) - 1\| \le C\delta_{\mathcal{R}}(U), \qquad \delta_{\mathcal{R}}(U) = \max_{r\in\mathcal{R}} \|r(U) - 1\|\,. \tag{4}$$

Here an empty maximum is zero. More precisely, if a finite set $\mathcal{R} \subseteq N$ and $\eta > 0$ satisfy

$$\delta_{\mathcal{R}}(U) < \eta \quad \Longrightarrow \quad \|w(U) - 1\| < 1 \tag{5}$$

in every dimension, then the same $\mathcal{R}$ satisfies (4) with $C = \pi/\eta$.

Proof. Suppose every MF homomorphism kills w. Exhaust N by finite increasing sets $\mathcal{R}_n$. If no threshold (5) exists, choose unitary tuples $U^{(n)}$ with $\delta_{\mathcal{R}_n}(U^{(n)}) < 1/n$ and $\left\|w(U^{(n)}) - 1\right\| \ge 1$. Every fixed relation has defect tending to zero, so the generator classes define a corona homomorphism of G whose value on w is not the identity. This contradicts Lemma 2.1.

Fix a threshold $(\mathcal{R}, \eta)$ and a tuple U, and put $t = \|w(U) - 1\|$. If $t = 0$, the required inequality is immediate. Otherwise Lemma 2.9 supplies $1 \le k \le \pi/t$ with $\left\|w(U)^{\otimes k} - 1\right\| > 1$. The map $A \mapsto A^{\otimes k}$ is a group homomorphism, so every word r satisfies

$$r(U_1^{\otimes k}, \dots, U_m^{\otimes k}) = r(U)^{\otimes k}.$$

The amplified tuple therefore violates the conclusion of (5), while its relation defect is at most $k\delta_{\mathcal{R}}(U)$. It follows that

$$\eta \le \delta_{\mathcal{R}}(U^{\otimes k}) \le k\delta_{\mathcal{R}}(U) \le \frac{\pi}{t}\delta_{\mathcal{R}}(U),$$

which proves (4).

Conversely, lift the generator images of a corona homomorphism to unitary matrices, as in Lemma 2.2. Their defects on the finite set $\mathcal{R}$ tend to zero, so (4) makes the displacement of w tend to zero. The corona homomorphism kills w, and Lemma 2.1 gives the assertion for MF targets. □

The same finite certificate kills w in every group satisfying the relations in $\mathcal{R}$. In particular, w is killed by every MF homomorphism of the finitely presented group $\langle x_1, \dots, x_m \mid \mathcal{R}\rangle$, which maps onto G. If $w \ne 1$ in G, its image in this finitely presented group is also nontrivial, so that group is non-MF. Thus the condition that a fixed word is killed by every MF homomorphism is open in the space of marked groups: each instance has an entire clopen relation cylinder on which it holds.

If G is given by a finite presentation with relator set $\mathcal{S}$, the inequality can use $\mathcal{S}$ itself. Express each $r \in \mathcal{R}$ as a product of a_r conjugates of elements of $\mathcal{S}$ and their inverses. Unitary conjugation preserves the norm, so telescoping gives $\|r(U) - 1\| \le a_r\delta_{\mathcal{S}}(U)$. Hence (4) holds with $\mathcal{S}$ and constant $C\max(1, \max_{r\in\mathcal{R}} a_r)$.

Taking the generator words gives the following consequence.

Proposition 2.11 (linear collapse). *Let $G = \langle g_1, \dots, g_m \rangle$ be a finitely generated group such that every homomorphism from G to an MF group is trivial. Then there are finitely many words $r_1, \dots, r_s$ in the free group F_m with $r_j(g_1, \dots, g_m) = 1$ in G, and a constant C, such that all unitaries $U_1, \dots, U_m \in \mathcal{U}(d)$, for every $d \geq 1$, satisfy*

$$\max_i \|U_i - 1\| \leq C \max_j \|r_j(U_1, \dots, U_m) - 1\| .$$

Conversely, if such words and such a constant exist, then every homomorphism from G to an MF group is trivial.

Proof. Apply Theorem 2.10 to each generator word, take the union of the resulting finite relation sets, and take the maximum of their positive constants. Conversely, the displayed inequality and the converse of that theorem kill every generator in every homomorphism to an MF group. □

With $r_1, \dots, r_s$ as in Proposition 2.11, put $P = \langle x_1, \dots, x_m \mid r_1, \dots, r_s \rangle$. The inequality holds verbatim for P, so every finitely generated group all of whose homomorphisms to MF groups are trivial is a quotient of a finitely presented group with the same property and the same number of generators. The set of m-marked groups satisfying $r_1 = \cdots = r_s = 1$ is open and closed in the space of marked groups, and every group in it has the property; so the property is open. For the two-generated group Q of Theorem D, the inequality bounds $\max\{\|U - 1\|, \|V - 1\|\}$ by the defects of finitely many words in two unitary matrices, uniformly over the matrix size, while the left regular representation solves the same equations exactly with $U, V \neq 1$. Applied to $\mathrm{EL}_4(\mathcal{C})$ from Theorem B, the same construction gives one finitely presented group with the property that maps into every $\mathrm{EL}_n(R)$ of that theorem with normally generating image.

3. One-sided inverses and elementary groups

The proof of Theorem B runs in rank four over the ring $\mathcal{C}$, with three coordinates for the subgroup $\mathrm{EL}_3(\mathcal{C})$, which has property (T) because $\mathcal{C}$ is finitely generated, and one for its centralizer; two lemmas then carry the conclusion from $\mathrm{EL}_4(\mathcal{C})$ to every $\mathrm{EL}_n(R)$. The compressor needs only $ts = 1$; the ideal condition on $1 - st$ enters in the last step of the rank-four argument and in the first lemma.

Lemma 3.1 (two copies). *Let R be a unital ring with $s, t \in R$ such that $ts = 1$ and $R(1 - st)R = R$. Then there are $v_0, v_1, w_0, w_1 \in R$ with $w_i v_j = \delta_{ij}$. Conversely, such elements satisfy $w_0 v_0 = 1$ and $w_1(1 - v_0 w_0)v_1 = 1$.*

Proof. Put $e = 1 - st$, so that $es = te = 0$, and choose $a_j, b_j \in R$, $0 \leq j < m$, with $\sum_j a_j e b_j = 1$. Put

$$v_0 = s^m, \qquad w_0 = t^m, \qquad v_1 = \sum_j s^j e b_j, \qquad w_1 = \sum_j a_j e t^j.$$

For all $i, j \geq 0$, $et^i s^j e = \delta_{ij} e$: for $j > i$ the middle factor is s^{j-i}, which e kills on the left, and for $i > j$ it is t^{i-j}, which e kills on the right. So $w_0 v_0 = t^m s^m = 1$, $w_1 v_1 = \sum_j a_j e b_j = 1$, $w_0 v_1 = \sum_j t^{m-j} e b_j = 0$, and $w_1 v_0 = \sum_j a_j e s^{m-j} = 0$. Conversely, $w_1(1 - v_0 w_0)v_1 = w_1 v_1 - (w_1 v_0)(w_0 v_1) = 1$. □

Lemma 3.2 (normal generation in rank two)**.** *Let R be a unital ring and let $v, w, a, b \in R$ satisfy $wv = 1$, $ba = 1$, and $bv = 0$. Then*

$$D = [e_{12}(v), e_{21}(b)] = \mathrm{diag}(1 + vb, 1)$$

normally generates $\mathrm{EL}_2(R)$*.*

Proof. The matrix $e_{12}(v)e_{21}(b)e_{12}(-v)$ has rows $(1+vb,\ -vbv)$ and $(b,\ 1-bv)$, so for $bv = 0$ it is $\left(\begin{smallmatrix} 1+vb & 0 \\ b & 1 \end{smallmatrix}\right)$, and multiplying by $e_{21}(-b)$ gives $D = \mathrm{diag}(1+vb, 1)$. Let N be the normal closure of D in $\mathrm{EL}_2(R)$. For $r \in R$, $[D, e_{12}(ar)] = e_{12}((1+vb)ar - ar) = e_{12}(vr)$, since $ba = 1$; so $e_{12}(vR) \leq N$. Put $f = 1 - vw$, an idempotent with $fv = 0 = wf$, and

$$z = vf + fw + 1 - f - vfw.$$

Put $x = vf$, $y = fw$, $q = vfw$, and $c = 1 - f - q$, so that $z = x + y + c$. From $wv = 1$ and $fv = wf = 0$: $x^2 = y^2 = 0$, $xy = q$, $yx = f$, the idempotents f and q are orthogonal, and c annihilates x and y on both sides; so $z^2 = q + f + c = 1$ and $zf = xf + yf + cf = vf$. The factorization

$$\mathrm{diag}(z, z^{-1}) = e_{12}(z)e_{21}(-z^{-1})e_{12}(z)\, e_{12}(-1)e_{21}(1)e_{12}(-1)$$

puts $h = \mathrm{diag}(z, z)$ in $\mathrm{EL}_2(R)$, and $he_{12}(fr)h^{-1} = e_{12}(zfrz) = e_{12}(v\,frz) \in N$. So $e_{12}(r) = e_{12}(vwr)\, e_{12}(fr) \in N$ for every $r \in R$, and conjugating by $e_{12}(1)e_{21}(-1)e_{12}(1)$ gives $e_{21}(R) \leq N$. □

Lemma 3.3 (a rank-four compression cell)**.** *Let R be a unital ring, let $s, t \in R$ satisfy $ts = 1$, and put $e = 1 - st$. In $G = \mathrm{EL}_4(R)$ let $L = \mathrm{EL}_3(R)$ occupy coordinates $1, 2, 3$. There are $u \in \mathrm{Comp}_G(L)$ and $c \in C_G(L)$ with*

$$ucu^{-1} = e_{12}(e), \qquad [ucu^{-1}, e_{23}(1)] = e_{13}(e).$$

Every off-diagonal entry of a matrix in uLu^{-1} has the form sat with $a \in R$.

Proof. Here $e^2 = e$ and $es = te = 0$. For $i = 1, 2, 3$, set

$$u_i = e_{4i}(t-1)e_{i4}(1)e_{4i}(s-1)e_{i4}(-t).$$

Its block on coordinates $(i, 4)$ is $\left(\begin{smallmatrix} s & e \\ 0 & t \end{smallmatrix}\right)$, so $u = u_3u_2u_1 \in \mathrm{EL}_4(R)$ is the matrix

$$u = \begin{pmatrix} s & 0 & 0 & e \\ 0 & s & 0 & et \\ 0 & 0 & s & et^2 \\ 0 & 0 & 0 & t^3 \end{pmatrix},$$

invertible as a product of elementary matrices. For $1 \leq i \neq j \leq 3$ and $a \in R$,

$$u\, e_{ij}(a) = e_{ij}(sat)\, u : \tag{6}$$

both sides differ from u by one entry, sa in position (i, j), since on the right the increment $sat \cdot (s, \ldots, et^{j-1})$ from row j of u contributes $sa\,ts = sa$ in position (i, j) and $sa\,te\,t^{j-1} = 0$ in position $(i, 4)$. So $uLu^{-1} \leq L$, and (6) gives the last assertion.

The element

$$c = [e_{41}(e), e_{14}(t)] = \mathrm{diag}(1, 1, 1, 1 + et)$$

is computed in the block on coordinates $(1, 4)$ using $te = 0$, and a diagonal matrix of this shape commutes with every $\mathrm{diag}(A, 1)$, so $c \in C_G(L)$. Both uc and $e_{12}(e)u$

equal $u + et\, E_{14}$: the last column of uc is $(e, et, et^2, t^3)^\mathsf{T}(1 + et)$ and $e^2 = e$, $te = 0$, while $e_{12}(e)u = u + e\,(0, s, 0, et)E_{1*}$ and $es = 0$. So $ucu^{-1} = e_{12}(e)$, and the Steinberg relation $[e_{12}(e), e_{23}(1)] = e_{13}(e)$ gives the second identity. □

Proof of Theorem B. *The universal group.* In $\mathcal{C}$ put $s = s_0$, $t = t_0$, and $e = 1 - st$. Then $ts = 1$, $e^2 = e$, $es = te = 0$, and $t_1 e s_1 = 1$. Put $G = \mathrm{EL}_4(\mathcal{C})$ and $L = \mathrm{EL}_3(\mathcal{C})$ on coordinates $1, 2, 3$. The ring $\mathcal{C}$ is nonzero, since it maps onto $L_{\mathbb{F}_2}(1, 2)$, and finitely generated, so both groups have property (T) by [EJZ, Theorem 1.1], and $\mathrm{EL}_4(\mathcal{C})$ is finitely generated [BHV, Theorem 1.3.1].

By Lemma 3.3 there are $u \in \mathrm{Comp}_G(L)$ and $c \in C_G(L)$ with

$$d = [ucu^{-1}, e_{23}(1)] = e_{13}(e) \in \mathfrak{D}_G(L).$$

Let N be the normal closure of d in G. For arbitrary $a, b \in \mathcal{C}$, the Steinberg relations give

$$[e_{41}(a), e_{13}(e)] = e_{43}(ae), \qquad [e_{43}(ae), e_{32}(b)] = e_{42}(aeb).$$

So $e_{42}(1) = e_{42}(t_1 e s_1) \in N$. Elementary signed permutation matrices conjugate this element to every off-diagonal position, up to a sign, so $e_{ij}(1) \in N$ whenever $i \neq j$. For distinct i, j, k and every $r \in \mathcal{C}$,

$$[e_{ij}(1), e_{jk}(r)] = e_{ik}(r) \in N.$$

So $N = G$, and $\mathfrak{D}_G(L) = G$. By Theorem A with $K = G$, every homomorphism from G to an MF group is trivial.

Rank two. Now let R satisfy the hypothesis. By Lemma 3.1 choose $v_0, v_1, w_0, w_1 \in R$ with $w_i v_j = \delta_{ij}$. Then $\varphi\colon s_i \mapsto v_i$, $t_i \mapsto w_i$ is a unital ring homomorphism $\mathcal{C} \to R$, and it induces a homomorphism $\mathrm{EL}_4(\mathcal{C}) \to \mathrm{EL}_4(R)$. Put

$$S_1 = v_0 v_0, \quad S_2 = v_0 v_1, \quad S_3 = v_1 v_0, \quad S_4 = v_1 v_1,$$
$$T_1 = w_0 w_0, \quad T_2 = w_1 w_0, \quad T_3 = w_0 w_1, \quad T_4 = w_1 w_1,$$

so that $T_i S_j = \delta_{ij}$, and put $p = \sum_i S_i T_i$. For a 4×4 matrix A over R put

$$\jmath(A) = 1 - p + \sum_{i,j} S_i A_{ij} T_j \in R.$$

Since $(1 - p)S_i = 0$ and $T_j(1 - p) = 0$, we have $\jmath(AB) = \jmath(A)\jmath(B)$ and $\jmath(I) = 1$, so $\Psi(A) = \mathrm{diag}(\jmath(A), 1)$ defines a homomorphism $\mathrm{GL}_4(R) \to \mathrm{GL}_2(R)$, and for $i \neq j$ and $r \in R$,

$$\Psi(e_{ij}(r)) = \mathrm{diag}(1 + S_i r T_j, 1) = [e_{12}(S_i r), e_{21}(T_j)]$$

by the computation in Lemma 3.2, since $T_j S_i = 0$. So $\Psi(\mathrm{EL}_4(R)) \leq \mathrm{EL}_2(R)$. Let ρ be a homomorphism from $\mathrm{EL}_2(R)$ to an MF group. Composed with $\mathrm{EL}_4(\mathcal{C}) \to \mathrm{EL}_4(R) \to \mathrm{EL}_2(R)$ it is trivial, so ρ kills the image $\Psi(e_{12}(1)) = \mathrm{diag}(1 + S_1 T_2, 1)$ of $e_{12}(1) \in \mathrm{EL}_4(\mathcal{C})$, which normally generates $\mathrm{EL}_2(R)$ by Lemma 3.2 with $v = S_1$, $w = T_1$, $a = S_2$, $b = T_2$. So ρ is trivial, and the image of $\mathrm{EL}_4(\mathcal{C})$ in $\mathrm{EL}_2(R)$ normally generates $\mathrm{EL}_2(R)$.

All ranks. For $n \geq 3$, signed permutation matrices conjugate $e_{12}(r)$ to $e_{ij}(\pm r)$ for all $i \neq j$, so the copy of $\mathrm{EL}_2(R)$ on coordinates $1, 2$ normally generates $\mathrm{EL}_n(R)$. A homomorphism from $\mathrm{EL}_n(R)$ to an MF group is trivial on that copy, hence trivial, and the image of $\mathrm{EL}_4(\mathcal{C})$ in that copy normally generates $\mathrm{EL}_n(R)$. □

Corollary 3.4. *If R is a countable simple unital ring that is not directly finite, then every homomorphism from $\mathrm{EL}_n(R)$ to an MF group is trivial for every $n \geq 2$. The same conclusion holds for $R = L_k(1, m)$, for every countable field k and every $m \geq 2$.*

Proof. In the first case choose $ts = 1 \neq st$; then $1 - st$ is a nonzero idempotent, and since R is simple it generates R as a two-sided ideal. In the second case, with generators $s_1, \dots, s_m, t_1, \dots, t_m$ subject to $t_i s_j = \delta_{ij}$ and $\sum_i s_i t_i = 1$, take $s = s_1$ and $t = t_1$; then $1 - s_1 t_1 = \sum_{i \geq 2} s_i t_i$ and

$$t_2(1 - s_1 t_1)s_2 = 1.$$

Theorem B applies in both cases. □

Corollary 3.5. *Let R be a countable unital ring. If R is not directly finite, then $C^*_{\max}(\mathrm{EL}_n(R))$ contains a proper isometry for every $n \geq 4$. If $R \neq 0$ satisfies the hypothesis of Theorem B, then for every $n \geq 2$ the algebra $C^*_{\max}(\mathrm{EL}_n(R))$ contains a proper isometry and $C^*_{\mathrm{r}}(\mathrm{EL}_n(R))$ is separable, stably finite, and not MF, and the unit group $R^\times$ is not MF.*

Proof. Choose $s, t \in R$ with $ts = 1 \neq st$, and let S be the unital subring they generate. Let u and $L = \mathrm{EL}_3(S)$ be as in Lemma 3.3 for S; then L has property (T) [EJZ, Theorem 1.1], and the compression is strict, since $e(sat) = 0$ while $e \cdot 1 = e \neq 0$ puts $e_{12}(1)$ in $L \setminus uLu^{-1}$. By Proposition 2.8 applied to $L \leq \mathrm{EL}_n(R)$ and u, $C^*_{\max}(\mathrm{EL}_n(R))$ contains a proper isometry. If $R \neq 0$ satisfies the hypothesis of Theorem B, take instead $S = \varphi(\mathcal{C})$, the map Ψ from the rank-two part of its proof, and u, $L = \mathrm{EL}_3(S)$ from Lemma 3.3 for S; here $v_0 w_0 \neq 1$, since $w_1 v_0 = 0$ and $w_1 v_1 = 1$. Then Ψ is injective, since $T_i \jmath(A) S_j = A_{ij}$, so $\Psi(L) \leq \mathrm{EL}_2(R) \leq \mathrm{EL}_n(R)$ has property (T), as S is finitely generated, and is compressed strictly by $\Psi(u)$, and Proposition 2.8 gives the proper isometry for every $n \geq 2$.

For the remaining assertions, $G = \mathrm{EL}_n(R)$ is countable and nontrivial, and by Theorem B it is not MF. The algebra $C^*_{\mathrm{r}}(G)$ is separable, stably finite because its canonical trace is faithful, and not MF by Lemma 2.1. Finally, $\jmath$ embeds $\mathrm{EL}_4(S)$ in $R^\times$; the group $\mathrm{EL}_4(S)$ is nontrivial with every homomorphism to an MF group trivial, so it is not MF, and restricting the maps V_n shows that MF passes to subgroups. So $R^\times$ is not MF. □

3.1. **The binary example.** Put $R = L_{\mathbb{F}_2}(1, 2)$ and $H = R^\times$. By (3), the maps $x \mapsto (t_0 x, t_1 x)$ and $(y, z) \mapsto s_0 y + s_1 z$ are mutually inverse isomorphisms of right R-modules between R and $R \oplus R$, and $H \cong \mathrm{GL}_4(R) = \mathrm{EL}_4(R)$ [K–T, Proposition 4.2 and Corollary 4.4]. We identify H with $\mathrm{EL}_4(R)$.

Proof of Theorem C. The relations give $t_0 s_0 = 1$ and $t_1(1 - s_0 t_0)s_1 = 1$, so R satisfies the hypothesis of Theorem B, and every homomorphism from H to an MF group is trivial. The ring R is finitely generated, so H has property (T) by [EJZ, Theorem 1.1], and hence is finitely generated [BHV, Theorem 1.3.1]. Since $e_{12}(1) \neq 1$, the group H is nontrivial.

Let N be a normal subgroup of H. The ring R is purely infinite simple [AA] and therefore an exchange ring [Ara], so by Preusser's sandwich theorem [Pre, Theorem 3] there is an ideal I of R with

$$\mathrm{EL}_4(R,I) \le N \le C_4(R,I),$$

where $\mathrm{EL}_4(R,I)$ is the normal closure in $\mathrm{EL}_4(R)$ of the elementary matrices $e_{ij}(a)$ with $a \in I$, and $C_4(R,I)$ is the preimage of the center of $\mathrm{GL}_4(R/I)$. Since R is simple, either $I = R$, and then $N \ge \mathrm{EL}_4(R) = H$, or $I = 0$, and then every $g \in N$ is central in $\mathrm{GL}_4(R)$. In the second case, commuting with each $e_{ij}(1)$ forces $g = \lambda I_4$ with $\lambda \in R^\times$, and commuting with each $e_{ij}(a)$ then gives $\lambda a = a\lambda$ for all $a \in R$. So λ lies in $Z(R) = \mathbb{F}_2$ [AC, Corollary 4.3], and $N = 1$. So H is simple, and in particular not MF.

Since $R \ne 0$ satisfies the hypothesis of Theorem B, Corollary 3.5 shows that $C^*_{\mathrm{r}}(H)$ is separable, stably finite, and not MF, and that $C^*_{\max}(H)$ contains a proper isometry. $\square$

All rings in the following discussion are unital and associative. Two idempotents e, f of a ring are *equivalent* if $e = xy$ and $f = yx$ for some x, y; an idempotent e is *infinite* if $e = f + g$ for orthogonal idempotents f, g with f equivalent to e and $g \ne 0$; and a simple ring is *purely infinite* if every nonzero right ideal contains an infinite idempotent [AGP, Definitions 1.2]. The algebras $L_k(1,d)$ are purely infinite simple [AA]. Let $\mathrm{GL}(R)$ and $\mathrm{EL}(R)$ be the direct limits of $\mathrm{GL}_n(R)$ and $\mathrm{EL}_n(R)$ along $A \mapsto \mathrm{diag}(A,1)$; the subgroup $\mathrm{EL}(R)$ is normal in $\mathrm{GL}(R)$ by Whitehead's lemma, and $K_1(R) = \mathrm{GL}(R)/\mathrm{EL}(R)$. Write $\kappa\colon R^\times \to K_1(R)$ for the canonical map.

Theorem 3.6 (MF quotients of unit groups)**.** *Let R be a countable unital purely infinite simple ring and let $n \ge 1$. Then every homomorphism from $\mathrm{GL}_n(R)$ to an MF group factors uniquely through the canonical map $\mathrm{GL}_n(R) \to K_1(R)$, and $K_1(R)$, a countable abelian group, is MF. Equivalently, the intersection N_n of the kernels of all homomorphisms from $\mathrm{GL}_n(R)$ to MF groups is $[\mathrm{GL}_n(R), \mathrm{GL}_n(R)]$, and $\mathrm{GL}_n(R)/N_n \cong K_1(R)$.*

Proof. The ring $M_n(R)$ is again countable, purely infinite, and simple [AGP, Corollary 1.7], and $K_1(M_n(R)) \cong K_1(R)$ by Morita invariance, so it suffices to treat $n = 1$. Write $H = R^\times$ and $N = N_1$. Ara, Goodearl, and Pardo show that κ is surjective with kernel $[H,H]$ [AGP, Theorem 2.4]. A countable abelian group A is MF: $C^*_{\max}(A)$ is commutative and separable, hence residually finite-dimensional and MF [BK], and Lemma 2.1 applies. So κ is a homomorphism to an MF group, and $N \le \ker \kappa$.

For the converse we use Theorem B in the following form. Every nonzero idempotent e of R is infinite [AGP, Proposition 1.5], so eRe contains s, t with $ts = e \ne st$, and eRe is simple, so $e - st$ generates eRe as a two-sided ideal; hence every homomorphism from $\mathrm{EL}_m(eRe)$, $m \ge 2$, to an MF group is trivial. Consequently, if e and P are nonzero idempotents and $\theta\colon M_m(eRe) \to PRP$ is a ring isomorphism, then

$$1 - P + \theta(A) \in N \qquad (A \in \mathrm{EL}_m(eRe)), \tag{7}$$

since these units form the image of $\mathrm{EL}_m(eRe)$ under a homomorphism to H.

We use two facts from the proof of [AGP, Theorem 2.4], which rest on [AGP, Proposition 1.5 and Corollary 1.7] and on the elementary reduction of Menal and Moncasi [MM, proof of Theorem 2.2, and the Remark after Corollary 2.3].

(a) There are orthogonal idempotents $e_1, \dots, e_m$ with sum 1, $m \ge 4$, such that $e_1, \dots, e_{m-1}$ are pairwise equivalent and e_m is equivalent to an idempotent

$f \le e_1$. Then R is the ring of $m \times m$ matrices over $T = e_1 R e_1$ whose last column has entries in Tf and whose last row has entries in fT, and every unit u of R factors as $u = PvQ$ with P and Q products of elementary matrices $e_{ij}(x)$ of this matrix ring and $v = e_1 + (1-e_1)v(1-e_1)$.

(b) For a nonzero idempotent e and $n \ge 2$ there are an idempotent $f < e$ equivalent to e and orthogonal idempotents $r_2, \dots, r_n \le e - f$, each equivalent to 1, such that, with $r_1 = 1 - e + f$ and $P = r_1 + \dots + r_n$, some ring isomorphism $\theta\colon M_n(R) \to PRP$ sends $\operatorname{diag}(v, 1, \dots, 1)$ to $(1-e)v(1-e) + f + r_2 + \dots + r_n$ for every unit $v = e + (1-e)v(1-e)$.

Let $u \in \ker\kappa$ and write $u = PvQ$ as in (a). The elementary matrices with $i, j \le m-1$ form $\mathrm{EL}_{m-1}(T)$ inside $(1-e_m)R(1-e_m) \cong M_{m-1}(T)$, so they lie in N by (7); for distinct $i, j \le m-1$,

$$e_{im}(x) = [e_{ij}(1), e_{jm}(x)], \qquad e_{mj}(x) = [e_{mi}(x), e_{ij}(1)]$$

are products of an element of $\mathrm{EL}_{m-1}(T)$ and a conjugate of its inverse, so they lie in the normal subgroup N as well. Hence $u \equiv v$ modulo N, and $\kappa(v) = \kappa(u) = 0$ because $N \le \ker\kappa$.

So let $v = e + (1-e)v(1-e)$ with $e \ne 0$ and $\kappa(v) = 0$. By the definition of K_1, $\operatorname{diag}(v, 1, \dots, 1) \in \mathrm{EL}_n(R)$ for some $n \ge 2$. With θ and P as in (b), $v = 1 - P + \theta(\operatorname{diag}(v, 1, \dots, 1))$, which lies in N by (7). Hence $\ker\kappa \le N$. $\square$

Corollary 3.7 (Leavitt unit groups)**.** *Let k be a countable field, let $d \ge 2$, let $R = L_k(1,d)$, and let $H = R^\times$. Then $H \cong \mathrm{GL}_d(R)$, the intersection of the kernels of all homomorphisms from H to MF groups is $[H,H] = \mathrm{EL}_d(R)$, and*

$$H/\mathrm{EL}_d(R) \cong K_1(R) \cong k^\times/(k^\times)^{d-1}.$$

Proof. The maps $x \mapsto (t_1 x, \dots, t_d x)$ and $(y_1, \dots, y_d) \mapsto \sum_i s_i y_i$ are mutually inverse isomorphisms of right R-modules between R and R^d, so $R \cong M_d(R)$ and $H \cong \mathrm{GL}_d(R)$. Theorem 3.6 identifies the intersection with $[H,H]$ and the quotient with $K_1(R)$. Khanh–Thanh show that $\mathrm{GL}_d(R) = \mathrm{EL}_d(R) D_d(k)$, where $D_d(k)$ is the abelian group of diagonal matrices with entries in $k^\times$, which normalizes $\mathrm{EL}_d(R)$, and that $K_1(R) \cong k^\times/(k^\times)^{d-1}$ [K–T, proof of Theorem 7.2]. So $[H,H] \le \mathrm{EL}_d(R)$, while $\mathrm{EL}_d(R) \le [H,H]$ because every homomorphism from $\mathrm{EL}_d(R)$ to an MF group is trivial (Theorem B). $\square$

For $d = 2$ the quotient is trivial for every countable field k, so every homomorphism from $L_k(1,2)^\times$ to an MF group is trivial. For $k = \mathbb{F}_q$ the quotient is cyclic of order $\gcd(q-1, d-1)$.

4. Amenable traces and Clifford lamps

Brown observed that it was unknown whether every amenable trace on a C^*-algebra is quasidiagonal [Br, discussion preceding Proposition 3.5.1]. For a separable unital C^*-algebra A we use the sequential form of his definitions. A tracial state τ on A is *amenable* if there are u.c.p. maps $\phi_n\colon A \to M_{d_n}(\mathbb{C})$ such that

$$\|\phi_n(ab) - \phi_n(a)\phi_n(b)\|_2 \longrightarrow 0, \qquad \mathrm{tr}_{d_n}(\phi_n(a)) \longrightarrow \tau(a) \quad (a, b \in A).$$

It is *quasidiagonal* if the first limit holds in operator norm instead of normalized Hilbert–Schmidt norm, with the same trace convergence.

For a countable group G, let u_g denote the canonical unitary in $C^*_{\max}(G)$. The canonical trace τ_G is determined by

$$\tau_G(u_g) = \begin{cases} 1, & g = 1, \\ 0, & g \neq 1. \end{cases}$$

Theorem 4.1 (canonical-trace obstruction)**.** *Let G be a countable group. If G is not MF, then its canonical trace τ_G on $C^*_{\max}(G)$ is not quasidiagonal. In particular, if τ_G is amenable, then τ_G is an amenable trace that is not quasidiagonal.*

Proof. Suppose that τ_G is quasidiagonal, with u.c.p. maps $\phi_n\colon C^*_{\max}(G) \to M_{d_n}(\mathbb{C})$ as in the definition. Asymptotic multiplicativity in operator norm makes $a \mapsto [(\phi_n(a))_n]$ a unital $*$-homomorphism $\Phi\colon C^*_{\max}(G) \to \mathcal{Q}_{\mathbf{d}}$. If $\Phi(u_g) = 1$, then $\|\phi_n(u_g) - 1\| \to 0$, so $\mathrm{tr}_{d_n}(\phi_n(u_g)) \to 1$, while $\mathrm{tr}_{d_n}(\phi_n(u_g)) \to \tau_G(u_g)$; so $g = 1$. Hence $g \mapsto \Phi(u_g)$ is a corona homomorphism injective on G, and G is MF, contrary to the hypothesis. $\square$

A group N is *locally residually finite* if each of its finitely generated subgroups is residually finite.

Proposition 4.2 (amenable extensions of locally residually finite groups)**.** *Let $1 \to N \to G \to A \to 1$ be an extension of countable groups with N locally residually finite and A amenable. Then the canonical trace of $C^*_{\max}(N)$ is quasidiagonal, and the canonical trace of $C^*_{\max}(G)$ is amenable.*

Proof. Write $g \mapsto \bar{g}$ for the quotient map $G \to A$, and fix a finite set $E \subseteq G$, a finite Følner set $F \subseteq A$, and a section $\sigma\colon A \to G$. For $g \in E$ and $x \in F$ put

$$b(g,x) = \sigma(\bar{g}x)^{-1} g \sigma(x) \in N.$$

The subgroup N_0 generated by these finitely many elements is residually finite; choose a finite quotient $\theta\colon N_0 \to Q$ with $\theta(b(g,x)) \neq 1$ whenever $b(g,x) \neq 1$, put $H = \ker\theta$, and choose representatives $r_q \in N_0$ of the cosets of H in N_0. The cosets $\sigma(x) r_q H$, $x \in F$, $q \in Q$, are pairwise distinct, since equality forces $x = x'$ in A and then $q = q'$; let T be their set, let P be the projection of $\ell^2(G/H)$ onto $\ell^2(T)$, and put $\Phi(b) = P\lambda(b)P|_{\ell^2(T)}$ for $b \in C^*_{\max}(G)$, where λ is the quasi-regular representation. This map is u.c.p., and for $g \in E$ and $\bar{g}x \in F$,

$$g\,\sigma(x) r_q H = \sigma(\bar{g}x)\, r_{\theta(b(g,x))q} H \in T.$$

So for $g, h \in E$ the identity $\Phi(u_{gh}) - \Phi(u_g)\Phi(u_h) = P\lambda(g)(1-P)\lambda(h)P$ and the rank of $(1-P)\lambda(h)P$, at most $|\{x \in F : \bar{h}x \notin F\}|\,|Q|$, give

$$\|\Phi(u_{gh}) - \Phi(u_g)\Phi(u_h)\|_2^2 \leq \frac{|\{x \in F : \bar{h}x \notin F\}|}{|F|}.$$

The normalized trace of $\Phi(u_g)$ is the fraction of points of T fixed by g, and for $g \in E \setminus \{1\}$ it is zero: if $\bar{g} \neq 1$, then g moves the F-coordinate of every point, and if $\bar{g} = 1$, then $g\sigma(x) r_q H = \sigma(x) r_{\theta(b(g,x))q} H$ with $b(g,x) = \sigma(x)^{-1} g \sigma(x) \neq 1$, so $\theta(b(g,x))q \neq q$. Taking Følner sets with $|\{x \in F : \bar{h}x \notin F\}|/|F| \to 0$ for the finitely many $h \in E$, along an exhaustion of G by finite sets E, the maps Φ are asymptotically multiplicative in normalized Hilbert–Schmidt norm and their traces converge to τ_G on the canonical unitaries; density and contractivity extend both limits to $C^*_{\max}(G)$.

For the canonical trace of $C^*_{\max}(N)$, run the same construction with N in place of G, $A = 1$, and $F = \{1\}$: no boundary term appears, so the maps Φ are multiplicative on E in operator norm, and their traces vanish on $E \setminus \{1\}$; the same limits give quasidiagonality. $\square$

We now construct the simplest group to which Theorem A applies: its intrinsic defect subgroup is generated by a single commutator and contains a central involution. Here Γ may be any countable property-(T) group with a proper self-embedding. Let Γ be a countable group with property (T), let $\alpha\colon \Gamma \to \Gamma$ be injective but not surjective, and choose $a \in \Gamma \setminus \alpha(\Gamma)$. Put

$$T_\alpha = \varinjlim(\Gamma \xrightarrow{\alpha} \Gamma \xrightarrow{\alpha} \cdots), \qquad V = T_\alpha \rtimes \langle t\rangle,$$

where $tgt^{-1} = \alpha(g)$ on the level-zero copy of Γ, so that V is the ascending HNN extension of Γ along α and $T_\alpha = \bigcup_{n\geq 0} t^{-n}\Gamma t^n$. Put $X = V/\Gamma$, the left-coset space. The Clifford lamp group $\mathrm{Cl}(X)$ is the group with generators ε and c_x, $x \in X$, and relations

$$\varepsilon^2 = c_x^2 = 1, \qquad [\varepsilon, c_x] = 1, \qquad [c_x, c_y] = \varepsilon \quad (x \neq y).$$

To see that $\varepsilon \neq 1$, fix a total order on X, let E be the $\mathbb{F}_2$-vector space of finitely supported functions $X \to \mathbb{F}_2$, and for $f, g \in E$ put $B(f,g) = \sum_{x>y} f(x)g(y)$. Since B is bilinear,

$$(a, f)(b, g) = \big(a + b + B(f, g),\ f + g\big)$$

is a group law on $\mathbb{F}_2 \times E$, in which $(1, 0)$ is a central involution, each $(0, \delta_x)$ is an involution, and for $x \neq y$ exactly one of $B(\delta_x, \delta_y)$ and $B(\delta_y, \delta_x)$ equals 1, so the commutator of $(0, \delta_x)$ and $(0, \delta_y)$ is $(1, 0)$. So $\varepsilon \mapsto (1, 0)$, $c_x \mapsto (0, \delta_x)$ defines a homomorphism $\mathrm{Cl}(X) \to \mathbb{F}_2 \times E$. The relations let us write every element of $\mathrm{Cl}(X)$ as

$$\varepsilon^a c_{x_1} \cdots c_{x_r}, \qquad a \in \mathbb{F}_2, \quad x_1 < \cdots < x_r,$$

and this word maps to $(a, \delta_{x_1} + \cdots + \delta_{x_r})$, so the expression is unique and the homomorphism is an isomorphism. In particular, for a finite subset $Y \subseteq X$, the subgroup generated by ε and the c_y with $y \in Y$ has order $2^{|Y|+1}$, so $\mathrm{Cl}(X)$ is countable and locally finite. Killing ε leaves $E \cong \bigoplus_X C_2$, so $\mathrm{Cl}(X)$ is a central extension of $\bigoplus_X C_2$ by C_2, and W below is a central extension of the permutational wreath product $C_2 \wr_X V$ by $\langle\varepsilon\rangle$. The relations are invariant under permutations of X, so every permutation of X induces an automorphism of $\mathrm{Cl}(X)$ that permutes the c_x accordingly and fixes ε. In this way V acts on $\mathrm{Cl}(X)$ through its action on X, and we put

$$W = \mathrm{Cl}(X) \rtimes V. \tag{8}$$

Proposition 4.3 (Clifford obstruction from a proper self-embedding)**.** *The group W is not MF. More precisely, every homomorphism from W to an MF group kills ε.*

Proof. Identify Γ with its level-zero copy in $V \leq W$, and let c be the lamp at the root coset $\Gamma \in X$; it centralizes Γ, which fixes that coset. Then tct^{-1} is the lamp at $t\Gamma$ and $a(tct^{-1})a^{-1}$ is the lamp at $at\Gamma$. These cosets are distinct, because $t\Gamma = at\Gamma$

would mean $a \in t\Gamma t^{-1} = \alpha(\Gamma)$; so $x = tct^{-1}$ and $y = axa^{-1}$ are distinct lamps, involutions with $[x,y] = \varepsilon \neq 1$. Then $d = [x,a] = xy$ and $d^2 = xyxy = [x,y] = \varepsilon$, and $d = [tct^{-1}, a]$ is a generator of $\mathfrak{D}_W(\Gamma)$, since $t \in \mathrm{Comp}_W(\Gamma)$, $c \in C_W(\Gamma)$, and $a \in \Gamma$. So $\langle\varepsilon\rangle \leq \mathfrak{D}_W(\Gamma)$. The subgroup $\langle\varepsilon\rangle$ is central, so normal, and it is finite, so it has property (T). So Theorem A applies to the countable group W with $L = \Gamma$ and $K = \langle\varepsilon\rangle$: every homomorphism from W to an MF group is trivial on ε. In particular, W is not MF. □

For a concrete instance, take

$$\bar{\Gamma} = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}) = \left\{ \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix} : A \in \mathrm{SL}_3(\mathbb{Z}),\ v \in \mathbb{Z}^3 \right\} \leq \mathrm{GL}_4(\mathbb{Z}).$$

It is residually finite, since reduction modulo a suitable integer separates any two distinct integral matrices, and it has property (T) [BHV, Example 1.7.4(i)]. Let

$$D = \mathrm{diag}(2,2,2,1), \qquad \alpha(g) = DgD^{-1},$$

so that $\alpha(v,A) = (2v,A)$. This is injective, and its image consists of the affine matrices whose translation coordinates are all even, so $[\bar{\Gamma} : \alpha(\bar{\Gamma})] = 8$. Translation a by the first standard basis vector lies outside $\alpha(\bar{\Gamma})$. So the preceding construction applies to $\bar{\Gamma}$, α and a.

The lamp group $\mathrm{Cl}(X)$ is locally finite, so this example locates $\mathrm{Rad}_{\mathrm{MF}}(W)$ only between $\langle\varepsilon\rangle$ and $\mathrm{Cl}(X)$. Section 5 keeps $\bar{\Gamma}$, α and a and replaces the lamps by perfect finite blocks. The resulting group is finitely presented, its MF radical is computed exactly, and Proposition 4.2 together with Theorem 4.1 then supplies the amenable nonquasidiagonal trace of Theorem E.

5. Finitely presented sofic groups with exact MF radical

The normal Kazhdan subgroup in Theorem A provides a central corner on which to upgrade Hilbert–Schmidt collapse. A finite-dimensional invariant algebra permits a different upgrade: correct its covariance first, then normalize the component orthogonal to the fixed vectors. This will allow finite perfect lamp groups whose normal closure in the ambient group is itself residually finite.

5.1. Finite-dimensional fixed points.

For a subalgebra A normalized by a unitary representation $\rho(L)$, write $A^L = \{a \in A : \rho(g)a\rho(g)^* = a \text{ for all } g \in L\}$.

Theorem 5.1 (finite-dimensional fixed-point transport)**.** *Let G be countable, let $L \leq G$ have property (T), and let $tLt^{-1} \leq L$. If $\rho\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is a homomorphism and $A \leq \mathcal{Q}_{\mathbf{d}}$ is a finite-dimensional unital C^*-subalgebra normalized by $\rho(L)$, then*

$$A^{tLt^{-1}} = A^L.$$

In particular, if a finite subgroup $F \leq G$ is normalized by L and $f \in F$ centralizes tLt^{-1}, every homomorphism from G to an MF group sends f to an element centralizing the image of L.

Proof. We first give the finite-dimensional lifting and correction used in the proof. A unital homomorphism from $A \cong \bigoplus_i M_{r_i}(\mathbb{C})$ to a norm matrix corona lifts, on a cofinite set of coordinates, to unital $*$-homomorphisms $\phi_n \colon A \to M_{d_n}(\mathbb{C})$. Indeed, lift its diagonal matrix units and round them successively to orthogonal projections, using the complement of their sum for the last one. The corrections tend to zero. In each summand, compress a lift of e_{j1} between the chosen projections p_j, p_1. Its products with its adjoint tend to p_j, p_1, so, on a common tail, its polar partial isometry v_j has precisely these final and initial projections. Then $v_j v_k^*$ are exact matrix units lifting e_{jk}. A summand may be zero at some coordinates; injectivity of ϕ_n is not required.

Suppose $\beta \in \mathrm{Aut}(A)$ and unitaries U_n satisfy $\mathrm{Ad}(U_n)\phi_n(a) - \phi_n(\beta(a)) \to 0$ in norm for every $a \in A$. Put $f_n = \mathrm{Ad}(U_n)\phi_n$ and $g_n = \phi_n\beta$. Finite dimensionality gives uniform convergence of $f_n - g_n$ on the unit ball. With normalized Haar measure on $\mathcal{U}(A)$, the matrices

$$T_n = \int_{\mathcal{U}(A)} g_n(u) f_n(u)^* \, du$$

satisfy $\|T_n - 1\| \to 0$ and $T_n f_n(a) = g_n(a) T_n$. The latter identity follows first for unitaries by left invariance of Haar measure and then for all a by linearity. The polar unitary $Z_n = T_n(T_n^* T_n)^{-1/2}$ has the same intertwining property and $\|Z_n - 1\| \to 0$. Thus $Z_n U_n$ implements β exactly on $\phi_n(A)$ while differing from U_n by a norm-null sequence.

Lift ρ to an operator norm asymptotic representation U_n of G. Choose a finite symmetric generating Kazhdan set S for L with constant $\kappa > 0$, and put $\beta_g = \mathrm{Ad}(\rho(g))|_A$. Apply the correction just proved to each $s \in S$, obtaining unitaries $V_n(s)$ with

$$V_n(s)\phi_n(a)V_n(s)^* = \phi_n(\beta_s(a)), \qquad \|V_n(s) - U_n(s)\| \longrightarrow 0.$$

On a common tail, $\ker \phi_n$ and the trace $\mathrm{tr}_{d_n} \phi_n$ are invariant under each β_s, hence under L. Therefore

$$\pi_n(g)\phi_n(a) = \phi_n(\beta_g(a))$$

is a genuine unitary representation of L on the finite-dimensional Hilbert space $A_n = \phi_n(A)$ with normalized Hilbert–Schmidt inner product. For a fixed $g \in L$, a word in S supplies a unitary $V_n(g)$ implementing $\pi_n(g)$ on A_n. Asymptotic multiplicativity gives $\|V_n(g) - U_n(g)\| \to 0$, so

$$\|\mathrm{Ad}(U_n(g))x - \pi_n(g)x\|_2 \le 2\,\|U_n(g) - V_n(g)\|\;\|x\|_2 \qquad (x \in A_n). \tag{9}$$

Put $L' = tLt^{-1}$ and suppose $a \in A^{L'} \setminus A^L$. Let P_n be the orthogonal projection in A_n onto its $\pi_n(L)$-fixed vectors. Infinitely many $b_n = \phi_n(a) - P_n\phi_n(a)$ are nonzero; otherwise every $\phi_n(\beta_g(a) - a)$ would eventually be zero, making a fixed by L in the corona. Retain those coordinates and set $x_n = b_n / \|b_n\|_2$. Then $\|x_n\|_2 = 1$, the vector x_n is orthogonal to the L-fixed subspace, and $\pi_n(h)x_n = x_n$ for $h \in L'$. Equation (9) gives

$$\|\mathrm{Ad}(U_n(h))x_n - x_n\|_2 \longrightarrow 0 \qquad (h \in L').$$

Consequently $y_n = U_n(t)^* x_n U_n(t)$ is asymptotically L-fixed. Theorem 2.4 transports y_n forward by t, so x_n is asymptotically L-fixed as well. By (9),

$$\max_{s\in S} \|\pi_n(s)x_n - x_n\|_2 \longrightarrow 0.$$

The Kazhdan inequality bounds this maximum below by κ, since x_n is a unit vector orthogonal to the fixed subspace. This contradiction proves $A^{L'} \subseteq A^L$; the reverse containment follows from $L' \leq L$.

For the last assertion use $A = C^*(\rho(F))$, which is finite dimensional and normalized by $\rho(L)$, and apply the result to $a = \rho(f)$. Homomorphisms to MF groups reduce to this case by Lemma 2.1. $\square$

Finite dimensionality of A cannot be relaxed, even to a commutative algebra with an eight-point spectrum of arcs.

Theorem 5.2 (the hypothesis is sharp)**.** *Let $\bar\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, let $\alpha(v, A) = (2v, A)$, and let V be the ascending HNN extension of $\bar\Gamma$ along α. There is an injective homomorphism σ of V into the unitary group of a norm matrix corona for which the following hold.*

(1) There is a unital commutative subalgebra $\mathcal{A} \cong C(T)$, normalized by $\sigma(\bar\Gamma)$, where T is a wedge of eight arcs, with

$$\mathcal{A}^{\alpha(\bar\Gamma)} \neq \mathcal{A}^{\bar\Gamma}.$$

The failure already occurs on the nine-dimensional operator system spanned by the unit and the eight generators.

(2) There is a finitely generated sofic MF group G containing V and an involution c centralizing $\bar\Gamma$ such that $f = tct^{-1}$ centralizes $\alpha(\bar\Gamma)$, the $\bar\Gamma$-orbit of f has eight elements and generates $\mathbb{Z}^7 \rtimes C_2$, and $\|[f, u] - 1\| = 2$ for the translation u by the first basis vector.

Proof. For odd q let $X_q = (\mathbb{Z}/q)^3$ and let σ_q be the permutation representation of V on $\ell^2(X_q)$ in which (v, A, k) acts by $x \mapsto 2^k Ax + v$; this is defined because 2 is invertible modulo q. A nonidentity element of V acts nontrivially for all large odd q, and a nonidentity permutation matrix is at distance at least $\sqrt{2}$ from the identity, so the classes $\sigma(g) = [\sigma_q(g)]_q$ define an injective homomorphism into the corona.

Let S consist of the translations by $\pm e_i$ and the elementary matrices $I \pm E_{ij}$, a generating set of $\bar\Gamma$, and let ℓ_q be the distance from 0 in the corresponding Schreier graph on X_q, so that $|\ell_q(sx) - \ell_q(x)| \leq 1$. Put

$$R_q = \Big\lfloor \tfrac{1}{4}\log_2 q \Big\rfloor, \qquad h_q(x) = \max\Big(1 - \frac{\ell_q(x)}{R_q}, 0\Big),$$

discarding the finitely many q with $R_q = 0$. Then $\|h_q \circ s^{-1} - h_q\| \leq R_q^{-1}$ for $s \in S$, so the class h of the multiplication operators by h_q commutes with $\sigma(\bar\Gamma)$. A word of length at most R in S carries 0 to the reduction of an integer vector of sup norm at most $2^R - 1$, since a shear at most doubles that norm and a translation raises it by one; so h_q is supported on reductions of integer vectors of sup norm at most $2^{R_q} - 1$.

For $b \in \{0,1\}^3$ put $h_{b,q}(x) = h_q(2^{-1}(x-b))$ and let h_b be the corresponding class, so that $h_0 = \sigma(t)h\sigma(t)^{-1}$ and h_b is the conjugate of h_0 by the translation by b. The eight supports are disjoint for large q: a common point gives $2(z-z') \equiv b'-b$ modulo q with z, z' integer vectors of sup norm at most $2^{R_q}-1$, and $4 \cdot 2^{R_q}+1 < q$ eventually, so the congruence is an equality in $\mathbb{Z}^3$, whose left side is even and whose right side has entries in $\{-1,0,1\}$; hence $b = b'$. The values of h_q are the numbers $1-j/R_q$ for $0 \le j \le R_q$, which become dense in $[0,1]$, so each h_b has spectrum $[0,1]$. Eight orthogonal positive elements with spectrum $[0,1]$ generate a commutative algebra whose spectrum is eight arcs joined at their common zero, which gives $\mathcal{A} \cong C(T)$.

Since h commutes with $\sigma(\bar{\Gamma})$, its conjugate h_0 commutes with $\sigma(t\bar{\Gamma}t^{-1}) = \sigma(\alpha(\bar{\Gamma}))$. The translation by e_1 carries h_0 to h_{e_1}, and these are orthogonal and nonzero, so $h_0 \notin \mathcal{A}^{\bar{\Gamma}}$. This proves (1), the relevant elements lying in the span of the unit and the eight generators. The $\bar{\Gamma}$-action on the eight generators is the affine action on $\mathbb{F}_2^3$, through $\bar{\Gamma} \to \bar{\Gamma}/\alpha(\bar{\Gamma})$.

For (2), double the dimensions, replace σ_q by $\sigma_q \otimes I_2$, and set $C_q(x) = R(\pi h_q(x))$ with $R(\theta) = \left(\begin{smallmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{smallmatrix}\right)$, a reflection, so that $C_q^2 = 1$. As $\|R(\theta) - R(\theta')\| = 2|\sin\frac{\theta-\theta'}{2}| \le |\theta-\theta'|$, the class c of the C_q is an involution commuting with $\sigma(\bar{\Gamma})$, and $f = \sigma(t)c\sigma(t)^{-1}$ has the eight conjugates f_b given by $R(\pi h_{b,q})$. At $x = 0$ the blocks of f_0 and f_{e_1} are $R(\pi)$ and $R(0)$, because $2^{-1}e_1$ lies outside the support of h_q; hence $\|f_0 - f_{e_1}\| = 2$, which is $\|[f,u]-1\|$ since f_0 is unitary and $ufu^{-1} = f_{e_1}$. Products of two of these reflections are commuting rotations, and f_0 inverts each of them, so the seven elements f_bf_0 generate a copy of $\mathbb{Z}^7$ on which f_0 acts by inversion: a relation among them restricts on the arc indexed by b to a rotation by $\pi n_b s$ for every $s \in [0,1]$, forcing $n_b = 0$, and no odd word in the reflections is trivial because its blocks are reflections. Thus the orbit generates $\mathbb{Z}^7 \rtimes C_2$.

Finally let $G = \langle V, c\rangle$ and let B_0 be generated by the V-conjugates of c. These are block-diagonal reflections, so every finitely generated subgroup of B_0 is virtually abelian, and $B_0 \cap V = 1$ because a permutation moving a fibre is at distance at least $\sqrt{2}$ from every block-diagonal unitary. Writing $T_\alpha = \bigcup_n t^{-n}\bar{\Gamma}t^n$, the group $\bar{\Gamma}$ is commensurated by V, so each $t^{-n}\bar{\Gamma}t^n$ has finite orbits on the conjugates of c; a finite subset of $B_0 \rtimes T_\alpha$ therefore lies in $M \rtimes t^{-n}\bar{\Gamma}t^n$ with M finitely generated virtually abelian and the action having finite image, and such a group is residually finite. So $B_0 \rtimes T_\alpha$ is locally residually finite, and $G = (B_0 \rtimes T_\alpha) \rtimes \mathbb{Z}$ is sofic by the permanence theorems used above. It is MF because it is given inside a norm matrix corona. $\square$

So a finite subgroup cannot be replaced in Theorem 5.1 by a finitely generated virtually abelian one, even when the element has order two and a finite orbit.

The order of correction and normalization matters: the error in (9) is proportional to the norm of x, so it remains small for the normalized vectors even when the original b_n have vanishing Hilbert–Schmidt mass.

5.2. **Perfect finite blocks.** Let Γ be finitely presented with property (T), let $\alpha\colon \Gamma \to \Gamma$ be injective of finite index $m > 1$, and suppose the ascending HNN extension

$$V = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma)\ (\gamma \in \Gamma)\rangle$$

is residually finite. For a finite group K put

$$\Omega = \Gamma/\alpha(\Gamma), \qquad F = K^{\Omega}, \qquad \Lambda = F \rtimes \Gamma,$$

where Γ permutes the m coordinates of F through its left action on Ω, and define

$$E_K = \langle \Lambda, t \mid t\gamma t^{-1} = \alpha(\gamma)\ (\gamma \in \Gamma)\rangle. \tag{10}$$

This is an HNN extension of Λ with associated subgroups Γ and $\alpha(\Gamma)$, both proper in Λ. Killing F leaves V, and $\langle \Gamma, t\rangle$ is a copy of V by Britton's lemma, so the quotient map splits. Write B_K for its kernel, the normal closure of F.

Lemma 5.3 (block structure)**.** *B_K is the free product of the conjugates of F, one for each coset in V/Γ:*

$$B_K \cong \mathop{\ast}_{v\Gamma \in V/\Gamma} F_v, \qquad F_v \cong K^m.$$

Proof. Let T be the Bass–Serre tree of (10), with vertices E_K/Λ and edges $E_K/\alpha(\Gamma)$, the edge $g\alpha(\Gamma)$ joining $g\Lambda$ to $gt\Lambda$. Since B_K is normal with quotient V, its orbits on vertices and on edges are V/Γ and $V/\alpha(\Gamma)$, and its stabilizers are $g(B_K \cap \Lambda)g^{-1} = gFg^{-1}$ at a vertex and $g(B_K \cap \alpha(\Gamma))g^{-1} = 1$ at an edge, because Γ meets B_K trivially. In the quotient graph the edge $\bar{v}\alpha(\Gamma)$ runs from $\bar{v}\Gamma$ to $\bar{v}t\Gamma$. That is precisely the incidence of the Bass–Serre tree of the ascending HNN extension V, whose vertices are V/Γ and whose edges are $V/\alpha(\Gamma)$, and the two graphs have the same vertex and edge sets because B_K is normal with quotient V. The quotient graph is therefore a tree, and a graph of groups with trivial edge groups over a tree has the displayed free product as its fundamental group, with no free factor. □

Theorem 5.4 (perfect finite-block radical)**.** *If K is a nontrivial finite perfect group, then E_K is finitely presented and sofic, and*

$$\mathrm{Rad}_{\mathrm{MF}}(E_K) = \bigcap_{\substack{N \trianglelefteq E_K \\ [E_K:N]<\infty}} N = B_K \neq 1.$$

The subgroup B_K is perfect and residually finite, and $E_K/B_K \cong V$. Also $E_K = J_K \rtimes \mathbb{Z}$ with J_K locally residually finite, and the canonical trace on $C^*_{\max}(E_K)$ is amenable and not quasidiagonal. If K is nonabelian simple, any nonidentity element of a single coordinate of F normally generates B_K in E_K.

Proof. The group Λ is finitely presented, since F is finite and Γ is finitely presented, and (10) adds one generator and finitely many relations; so E_K is finitely presented.

Let ρ be a corona homomorphism from E_K. The finite subgroup F is normalized by Γ, and the coordinate of F at the point $\alpha(\Gamma) \in \Omega$ is centralized by the stabilizer of that point, which is $\alpha(\Gamma) = t\Gamma t^{-1}$. By the last assertion of Theorem 5.1, applied with $L = \Gamma$, the image of that coordinate centralizes $\rho(\Gamma)$. For $k \in K$ and $\gamma \in \Gamma$ the conjugate $\gamma k \gamma^{-1}$ is the copy of k at $\gamma\alpha(\Gamma)$, so

$$\rho(k_\omega) = \rho(k_{\alpha(\Gamma)}) \qquad (\omega \in \Omega).$$

Since $m > 1$, two distinct coordinates of F commute, so this common image is an abelian image of K and is trivial by perfection. Thus ρ kills F and hence B_K, and all MF homomorphisms kill B_K by Lemma 2.1. Conversely the residually finite quotient

V is MF and separates every element outside B_K, which proves $\mathrm{Rad}_{\mathrm{MF}}(E_K) = B_K$. Finite groups are MF, and the finite quotients of V give the same two containments for the intersection of finite-index normal subgroups.

By Lemma 5.3, B_K is a free product of copies of K^m, so it is perfect, and every nontrivial element survives the retraction onto a free product of finitely many factors. A free product of finitely many finite groups is residually finite: mapping it onto the direct product of the factors gives a finite-index kernel meeting every conjugate factor trivially, which therefore acts freely on the free-product tree and is free; free groups are residually finite, and a group with a residually finite normal subgroup of finite index is residually finite. So B_K is residually finite.

Write $\Gamma_n = t^{-n}\Gamma t^n$ and $T_\alpha = \bigcup_{n\geq 0} \Gamma_n$, so that

$$V = T_\alpha \rtimes \mathbb{Z}, \qquad E_K = J_K \rtimes \mathbb{Z}, \qquad J_K = B_K \rtimes T_\alpha.$$

Each Γ_n has finite index in Γ_{n+1} because $[\Gamma : \alpha(\Gamma)] = m$, and conjugation by t carries Γ_{n+1} onto Γ_n; so Γ is commensurated by V and every Γ_n-orbit in V/Γ is finite. A finite subset of J_K uses finitely many blocks and lies in one Γ_n; saturating those blocks gives a finite Γ_n-invariant set $I \subseteq V/\Gamma$ and puts the subset in $M_I \rtimes \Gamma_n$ with $M_I = *_{v\Gamma\in I} K^m$ finite in the sense of finitely many factors. The action of Γ_n on M_I permutes the $m|I|$ labeled coordinates, so its kernel has finite index and centralizes M_I; the product of M_I with that kernel is a residually finite subgroup of finite index. So J_K is locally residually finite, and soficity follows from the directed-union and amenable-extension permanence theorems [ES, Theorem 1]. Proposition 4.2 makes the canonical trace amenable, and Theorem 4.1 makes it nonquasidiagonal, since $B_K \neq 1$ and E_K is not MF.

Finally, for nonabelian simple K the normal closure of a nonidentity k in one coordinate contains that coordinate, then all its V-conjugates, and hence B_K; the reverse containment is normality of B_K. □

5.3. **The centerless affine example.**

Proof of Theorem E. Take $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, $\alpha(v, A) = (2v, A)$, and $K = A_5$ in Theorem 5.4. This is the group $\bar{\Gamma}$ and the endomorphism of Section 4; it has property (T) and is finitely presented. For the latter fact one may use a finite presentation of $\mathrm{SL}_3(\mathbb{Z})$ and its action on the three commuting translation generators; an explicit twenty-relator presentation and its identification with this affine group are also proved in the repository at `LiteralBaseCompleteness.baseAffineEquiv`. Here $m = 8$, and the faithful matrix model for V displayed after Proposition 2.8 is residually finite by reduction modulo odd integers. The height map $V \to \mathbb{Z}$ vanishes on Γ, so V/Γ is countably infinite. Hence $B_{A_5} \cong *_{j\geq 1}(A_5)^8$, and all the claimed properties except centerlessness follow from the theorem.

In the affine matrix model, an element of $Z(V)$ commuting with all translations has $2^h A = I_3$. Taking determinants gives $h = 0$ and $A = I_3$. Its translation vector must then be fixed by every integral elementary matrix, so is zero. Thus $Z(V) = 1$. A central element of E_{A_5} consequently lies in B_{A_5}, whose center is trivial by the normal form for a free product of at least two nontrivial groups. Therefore $E = E_{A_5}$ is centerless. □

The radical in this example is an MF group in its own right, since it is residually finite. Every MF representation of the ambient finitely presented group nevertheless annihilates it. No central marked element or normal property-(T) subgroup is needed in this construction.

6. A sofic group with exactly cyclic MF quotients

The algebraic step is a general HNN construction. For a countable group G, write $\mathrm{Res}_{\mathrm{fin}}(G)$ for the intersection of its finite-index normal subgroups.

Proposition 6.1 (cyclic HNN absorption)**.** *Let E be countable and let $B \trianglelefteq E$ be annihilated by every homomorphism from E to an MF group. Suppose $r \in E$ and $d \in B$ have infinite order, the image of r normally generates E/B, and $\langle\langle d \rangle\rangle_E = B$. Put*

$$P = \langle E, s \mid srs^{-1} = d \rangle, \qquad \chi \colon P \longrightarrow \mathbb{Z}, \quad \chi(E) = 0, \quad \chi(s) = 1.$$

Then $N := \ker \chi = \langle\langle d \rangle\rangle_P$ satisfies

$$\mathrm{Rad}_{\mathrm{MF}}(N) = N, \qquad \mathrm{Rad}_{\mathrm{MF}}(P) = \mathrm{Res}_{\mathrm{fin}}(P) = N.$$

For every $R \trianglelefteq P$, one has $\mathrm{Rad}_{\mathrm{MF}}(P/R) = NR/R$; moreover P/R is MF if and only if $d \in R$, if and only if P/R is cyclic.

Proof. The infinite-order assumption makes $r^k \mapsto d^k$ an isomorphism of the two cyclic edge groups, so Britton's lemma embeds E in P. The map χ is well defined and split by $k \mapsto s^k$. Writing a word of total s-exponent zero as a product of conjugates of elements of E shows that

$$N = \langle E_i : i \in \mathbb{Z} \rangle, \qquad E_i = s^i E s^{-i}.$$

Use analogous notation for B_i, r_i, d_i. The defining relation gives

$$r_{i+1} = d_i \qquad (i \in \mathbb{Z}). \tag{11}$$

Let $f \colon N \to M$ be a homomorphism to an MF group. Its restriction to every $E_i \cong E$ kills B_i, and hence d_i. Equation (11) then kills r_{i+1}. Since r normally generates E/B, the restriction to E_{i+1} is trivial. This holds for every integer i, so $f = 1$ and $\mathrm{Rad}_{\mathrm{MF}}(N) = N$.

Every MF map out of P consequently kills N and factors uniquely through χ. Conversely, the quotient $\mathbb{Z}$ is MF and its finite cyclic quotients separate all elements outside N. Finite groups are MF, so the two asserted radicals of P equal N. Killing d kills B inside E and then kills r by the HNN relation. Normal generation of E/B kills the rest of E. The resulting presentation has the single generator s and no relators. Thus $P/\langle\langle d \rangle\rangle_P \cong \mathbb{Z}$ via χ, and $\langle\langle d \rangle\rangle_P = N$.

For an arbitrary normal R, the image NR/R of N has no nontrivial MF image. It is therefore contained in the MF radical of P/R. The quotient P/NR is cyclic and hence MF, giving the reverse inclusion. Since $N = \langle\langle d \rangle\rangle_P$, this radical is trivial precisely when $d \in R$. In that case P/R is cyclic; the converse holds because cyclic groups are MF. ∎

Proof of Theorem F. Use the concrete group $E = B \rtimes V$ from Theorem E, and write $(A_5)_0$ for one coordinate of its base block F. Choose

$$x = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \in \mathrm{SL}_3(\mathbb{Z}), \qquad a = (12)(34), \quad b = (123) \in (A_5)_0,$$

and set

$$r = tx^{-1}, \qquad d = atbt^{-1}, \qquad P = \langle E, s \mid s(tx^{-1})s^{-1} = atbt^{-1} \rangle. \tag{12}$$

Thus P is given by the finite presentation of E with one additional generator and one additional relator.

We first prove that r normally generates V. In the quotient by r, one has $t = x$. Since t commutes with $\mathrm{SL}_3(\mathbb{Z})$, the image of x is central in the image of that linear group. Writing $e_{ij} = I + E_{ij}$, conjugation by x cycles e_{12}, e_{31}, e_{23}, so their images are equal. The elementary identity $[e_{12}, e_{23}] = e_{13}$ kills e_{13}. Conjugating by determinant-one signed permutation matrices kills every elementary generator (inverting when necessary). Integral elementary row reduction shows that these generate $\mathrm{SL}_3(\mathbb{Z})$, so the linear group dies. Then $t = x$ dies. Each translation generator satisfies $tv_it^{-1} = v_i^2$; hence $v_i = v_i^2$ and it dies as well. These elements generate V, proving $\langle\langle r \rangle\rangle_V = V$.

Let $h\colon E \to \mathbb{Z}$ be the original height map, with $h(t) = 1$ and $h(B) = h(\Gamma) = 0$. Since $h(r) = 1$, the element r has infinite order. The two letters a and tbt^{-1} lie in different free-product blocks of B: the block labels Γ and $t\Gamma$ have different heights. Thus d is cyclically reduced of length two and d^n has reduced length $2n$ for $n \geq 1$. In particular d has infinite order. In the quotient $E/\langle\langle d \rangle\rangle_E$, the images of a and tbt^{-1} are inverse, with orders dividing 2 and 3. Both are therefore trivial. Since a normally generates B by Theorem E, this gives $\langle\langle d \rangle\rangle_E = B$.

Proposition 6.1 now gives all radical and quotient assertions, with $N \neq 1$ because E embeds in P. The kernel is perfect: its abelianization is MF, since characters separate points of a countable abelian group, whereas every MF image of N is trivial. Every finite quotient of P factors through χ, and every finite cyclic quotient of $\mathbb{Z}$ occurs. These identifications respect quotient maps, proving $\widehat{P} \cong \widehat{\mathbb{Z}}$.

The base E is sofic and the edge groups are infinite cyclic, so Collins–Dykema's HNN theorem [CD, Corollary 3.6] proves that P is sofic. Its hypothesis is monotileable amenability of the edge subgroup, supplied here by intervals tiling $\mathbb{Z}$. For $p \leq q$, write $N_{[p,q]} = \langle E_i : p \leq i \leq q \rangle$. HNN normal form identifies $N_{[p,q]}$ with the iterated amalgam of these vertex groups along (11), and gives

$$N_{[p,q+1]} \cong N_{[p,q]} *_{\langle d_q \rangle = \langle r_{q+1} \rangle} E_{q+1}.$$

The edge subgroup is proper in E_{q+1}, which contains A_5. Thus every $N_{[p,q]}$ is a proper subgroup of N. Every finite subset of N lies in such an interval, proving that N is not finitely generated.

Finally, the two associated cyclic subgroups have trivial intersection inside E. Indeed $r^k = d^\ell$ implies $k = 0$ by height and then $\ell = 0$ by the infinite order of d. Both are proper in E. The kernel of the Bass–Serre tree action is contained in the intersection of these two edge stabilizers, and is therefore trivial. Bryder–Ivanov–Omland's criterion [BIO, Proposition 4.12] for a nonascending HNN extension with finite intersection of the associated subgroups proves that P is a Powers group and that $C^*_{\mathrm{r}}(P)$ is simple with a unique trace. Moreover $s\langle r\rangle s^{-1} = \langle d\rangle$ makes $\langle r\rangle$ weakly malnormal in P, so Minasyan–Osin's HNN criterion [MO, Corollary 2.3] proves acylindrical hyperbolicity. An MF embedding of $C^*_{\mathrm{r}}(P)$ would restrict to an MF embedding of P, contradicting $N \neq 1$. Thus this reduced algebra is not MF. □

A nontrivial finite normal subgroup of P would make the average of its canonical unitaries a nontrivial central projection of the simple algebra $C^*_{\mathrm{r}}(P)$, so the finite radical of P is trivial, and $C^*_{\mathrm{r}}(P)$ has stable rank one by Gerasimova and Osin [GO, Theorem 1.1].

The canonical faithful trace makes $C^*_{\mathrm{r}}(P)$ stably finite. At the level of the maximal algebra, every unital $*$-homomorphism from $C^*_{\max}(P)$ to an MF C^*-algebra factors through the quotient $C^*(\mathbb{Z}) \cong C(\mathbb{T})$: restrict to the generating group unitaries and apply Theorem F. Thus the same group has a simple reduced algebra and a cyclic universal MF quotient.

6.1. **Prescribed MF quotients.** One amalgamation turns P into a device that installs a prescribed universal MF quotient on a group of one's choosing. The kernel it attaches is invisible to MF groups and to rational homology at the same time.

Lemma 6.2 (the invisible kernel is rationally acyclic)**.** *$H_j(N;\mathbb{Q}) = 0$ for every $j > 0$, and χ induces an isomorphism on rational homology.*

Proof. The group B is a free product of finite groups, so its rational homology vanishes in positive degrees, and the Lyndon–Hochschild–Serre spectral sequence for $B \to E \to V$ makes $E \to V$ a rational homology isomorphism. Write $L = \mathbb{Z}[1/2]^3$ for the translation subgroup of V, so that $V/L \cong \mathrm{SL}_3(\mathbb{Z}) \times \langle t\rangle$. Then $H_j(L;\mathbb{Q}) = \bigwedge^j \mathbb{Q}^3$, and t acts on it by 2^j; for $1 \leq j \leq 3$ the map $2^j - 1$ is invertible over $\mathbb{Q}$, so both the invariants and the coinvariants of $\langle t\rangle$ in $H_j(L;\mathbb{Q})$ vanish, and the spectral sequence leaves only the row $j = 0$. The positive-degree rational homology of $\mathrm{SL}_3(\mathbb{Z})$ vanishes [So], so $H_*(V;\mathbb{Q}) \cong H_*(\mathbb{Z};\mathbb{Q})$ through the height map.

For N, use the graph of groups of (11): vertex groups E_i indexed by $\mathbb{Z}$, infinite cyclic edge groups $\langle d_i\rangle = \langle r_{i+1}\rangle$, and a line as underlying graph. In the resulting Mayer–Vietoris sequence the vertex and edge groups have rational homology only in degrees 0 and 1. In degree one the edge generator $d_i = r_{i+1}$ has height 0 in E_i and height 1 in E_{i+1}, so the edge-to-vertex map is

$$\bigoplus_{i\in\mathbb{Z}} \mathbb{Q} \longrightarrow \bigoplus_{i\in\mathbb{Z}} \mathbb{Q}, \qquad e_i \longmapsto e_{i+1},$$

up to sign, an isomorphism; and in degree zero it is injective because the graph is a tree. Exactness gives $H_j(N;\mathbb{Q}) = 0$ for $j > 0$, and then the spectral sequence for $N \to P \to \mathbb{Z}$ gives the last assertion. □

Theorem 6.3 (prescribed MF quotients)**.** *Let A be a finitely presented sofic group and let $a \in A$ have infinite order. Amalgamate the infinite cyclic subgroups generated by s and a,*

$$G_A = P *_{\langle s\rangle = \langle a\rangle} A,$$

and let $q\colon G_A \to A$ be the identity on A and $p \mapsto a^{\chi(p)}$ on P. Then G_A is finitely presented and sofic, it contains P, and it is not MF. Moreover:

(1) composition with q is a bijection from the homomorphisms $A \to M$ to the homomorphisms $G_A \to M$, for every MF group M; so $\mathrm{Rad}_{\mathrm{MF}}(G_A) = q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(A))$, and A is the largest MF quotient of G_A whenever A is MF;

(2) $\ker q \cong *_{A/\langle a\rangle} N$ is nontrivial, perfect, rationally acyclic, and equal to its own MF radical;

(3) q induces an isomorphism of profinite completions, and $H_j(G_A; q^*M) \cong H_j(A; M)$ and $H^j(A; M) \cong H^j(G_A; q^*M)$ for every $\mathbb{Q}A$-module M and every j;

(4) G_A is acylindrically hyperbolic with trivial finite radical, and $C^*_{\mathrm{r}}(G_A)$ is simple with a unique tracial state, has stable rank one, and is stably finite but not MF.

Proof. If $A = \langle a\rangle$ then $G_A = P$ and everything has been proved, so assume $\langle a\rangle \neq A$. The amalgamated subgroups are infinite cyclic and proper in both factors, so normal form embeds P and A, and a presentation of G_A is the union of finite presentations of P and A together with $s = a$. Soficity is the amalgamation theorem of Collins and Dykema over the monotileably amenable group $\mathbb{Z}$ [CD, Theorem 3.4]. The retraction q is well defined because $\chi(s) = 1$, and it splits the inclusion of A.

(1) Let $f\colon G_A \to M$ with M an MF group. Its restriction to P factors through χ by Theorem F, so it is determined by $f(s) = f(a)$, and $f = (f|_A) \circ q$. Conversely every homomorphism from A arises this way, and q is surjective, so the correspondence is bijective. The radical formula and the statement about the largest MF quotient follow, as does G_A not being MF, since $P \leq G_A$ is not.

(2) In the Bass–Serre tree of the amalgam, $\ker q$ meets A trivially and meets P in $\ker \chi = N$, and it meets the edge group trivially. Its quotient graph is a star: one vertex of A-type with trivial stabilizer, one vertex of P-type for each coset in $A/\langle a\rangle$ with stabilizer a conjugate of N, and one edge from the centre to each of them. A star is a tree, so $\ker q$ is the free product of those conjugates. A homomorphism from a free product to an MF group is trivial as soon as it is trivial on each factor, so $\mathrm{Rad}_{\mathrm{MF}}(\ker q) = \ker q$ by Theorem F; the remaining properties come from Lemma 6.2, since rational homology of a free product is the sum of the rational homologies of its factors.

(3) Finite groups are MF, so (1) makes the finite quotients of G_A and of A correspond, which gives the profinite completions. For the rest, $\ker q$ acts trivially on q^*M and, being rationally acyclic, has $H_j(\ker q; q^*M) = H_j(\ker q; \mathbb{Q}) \otimes M = 0$ for $j > 0$; both Lyndon–Hochschild–Serre spectral sequences of $\ker q \to G_A \to A$ therefore collapse onto a single row.

(4) Pick a nonidentity element z of finite order in one coordinate A_5 of $B \leq P$. If $s^k = z s^\ell z^{-1}$ then $k = \ell$ by χ, and $k \neq 0$ would give $s^k z s^{-k} = z$, which Britton's lemma forbids in the HNN extension P because z lies in neither associated subgroup, both of which are infinite cyclic while z has finite order. So $\langle s\rangle \cap z\langle s\rangle z^{-1} = 1$ and $\langle s\rangle$ is weakly malnormal in G_A. Minasyan–Osin's amalgam criterion [MO, Corollary 2.2] then makes G_A either virtually cyclic or acylindrically hyperbolic, and the first is excluded because G_A contains P. A finite normal subgroup fixes a vertex of the Bass–Serre tree,

hence fixes the whole tree by minimality, hence lies in $\langle s\rangle \cap z\langle s\rangle z^{-1} = 1$. Simplicity and uniqueness of the trace now follow from [DGO, Theorem 2.35], stable rank one from [GO, Theorem 1.1], stable finiteness from the faithful canonical trace, and failure of MF from Lemma 2.1. □

Corollary 6.4. *For every $m \geq 1$ there is a finitely presented sofic group G_m that is not MF, whose largest MF quotient is $\mathbb{Z}^m$, whose profinite completion is $\widehat{\mathbb{Z}}^m$, and whose rational cohomology ring is that of the m-torus, and for which $C^*_{\mathrm{r}}(G_m)$ is simple with a unique tracial state, has stable rank one, and is stably finite but not MF.*

Proof. Take $A = \mathbb{Z}^m$ and a a basis vector in Theorem 6.3; the cohomology isomorphism of (3) respects cup products because it is induced by a group homomorphism. □

The finite certificate of Section 2.5 turns this into a statement about every matrix model of P.

Proposition 6.5 (approximate models of P are nearly cyclic)**.** *Write $P = \langle x_1, \dots, x_m, s \mid \mathcal{R}\rangle$ for the finite presentation obtained from (12) by expanding a finite presentation of E, so that $x_1, \dots, x_m$ generate E and $\chi(s) = 1$. There is a constant C, independent of the dimension, such that every tuple $(X_1, \dots, X_m, S) \in \mathcal{U}(d)^{m+1}$ satisfies*

$$\max_i \|X_i - 1\| \leq C \max_{r\in\mathcal{R}} \|r(X_1, \dots, X_m, S) - 1\|\,.$$

Moreover the distance from that tuple to the set of homomorphisms $P \to \mathcal{U}(d)$, measured by the maximum over the generators, is exactly $\max_i \|X_i - 1\|$.

Proof. Each x_i lies in N, so every homomorphism from P to an MF group kills it, and Theorem 2.10 supplies a finite set of relations of P and a linear bound for x_i in terms of their defects. Each of those relations is a product of conjugates of elements of $\mathcal{R}^{\pm 1}$, so bounding its defect by a multiple of $\max_{r\in\mathcal{R}} \|r(X) - 1\|$ changes only the constant; take the largest constant over i.

For the second assertion, a homomorphism $P \to \mathcal{U}(d)$ has MF image, so it kills every x_i and sends the generators to $(1, \dots, 1, S')$ for some unitary S'; conversely χ followed by $1 \mapsto S$ is such a homomorphism for every $S \in \mathcal{U}(d)$. The distance is therefore at least $\max_i \|X_i - 1\|$, and the tuple $(1, \dots, 1, S)$ attains it. □

So the approximate matrix representations of P are linearly close to honest representations, all of which factor through χ, while P itself is sofic and not MF.

7. A torsion-free finitely presented example

For an acylindrically hyperbolic group G, choose the generating set A provided by Hull [Hu, Theorem 3.12]. A subgroup is *suitable* if it acts non-elementarily on the Cayley graph $\mathrm{Cay}(G, A)$ and normalizes no nontrivial finite subgroup [Hu, Definition 1.4]. Hull states his small cancellation theorem [Hu, Theorem 7.1(a), (b), (c), and (e)] with injectivity on a ball of $\mathrm{Cay}(G, A)$; since every finite subset of G lies in some ball, the theorem takes the following form.

Theorem 7.1 (Hull's small cancellation theorem)**.** *Let G be acylindrically hyperbolic, let $N \leq G$ be suitable with respect to A, and let $g_1, \dots, g_m \in G$ and a finite subset $\Omega \subseteq G$ be given. Then there is a surjective homomorphism $\varphi\colon G \to Q$ such that Q is acylindrically hyperbolic, $\varphi|_\Omega$ is injective, $\varphi(g_i) \in \varphi(N)$ for all i, and every element of finite order in Q is the image of an element of the same order in G.*

Hull's proof treats $m = 1$ by passing to $G/\langle\langle r\rangle\rangle_G$ for one element r and the general case by induction on m [Hu, proof of Theorem 7.1], so $\ker\varphi$ is the normal closure of m elements and Q is finitely presented when G is.

Lemma 7.2 (saturation)**.** *Let G be finitely presented, torsion-free, and acylindrically hyperbolic, let $N \trianglelefteq G$ be nontrivial, and let $\Omega \subseteq G$ be finite. Then there is a surjective homomorphism $\varphi\colon G \to Q$ such that Q is two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $\varphi|_\Omega$ is injective, and $\varphi(N) = Q$.*

Proof. The subgroup N is normal, and it is infinite because G is torsion-free and $N \neq 1$, so it acts non-elementarily on $\mathrm{Cay}(G, A)$ by Osin [Os, Lemma 7.1]; since G is torsion-free, N normalizes no nontrivial finite subgroup, so N is suitable. By Hull [Hu, Corollary 5.7 and Lemma 5.8], N contains two elements h_1, h_2 such that $N_0 = \langle h_1, h_2\rangle$ is again suitable, possibly with respect to a larger generating set $A' \supseteq A$, and Theorem 7.1 holds for A' as well, since every finite subset of G lies in a ball of $\mathrm{Cay}(G, A')$. Apply Theorem 7.1 to N_0 with $g_1, \dots, g_m$ a finite generating set of G and with the set Ω. Then $Q = \langle\varphi(g_1), \dots, \varphi(g_m)\rangle \leq \varphi(N_0) \leq \varphi(N) \leq Q$, so $Q = \varphi(N_0) = \langle\varphi(h_1), \varphi(h_2)\rangle$ and $\varphi(N) = Q$. The quotient is torsion-free by Theorem 7.1 and finitely presented because $\ker\varphi$ is normally generated by m elements. □

Fournier-Facio constructs a finitely presented torsion-free group G_0 with property (T), a subgroup $\Gamma \leq G_0$ with property (T), an element $t \in G_0$ with $t\Gamma t^{-1} \leq \Gamma$, and a subgroup $J \leq G_0$ isomorphic to a finitely presented infinite simple group, such that $[\Gamma, J] = 1$ and $tJt^{-1} \leq \Gamma$ [FF, §2]. The group G_0 is obtained there as a common quotient of two finitely generated acylindrically hyperbolic groups by Hull's theorem [Hu, Corollary 7.4], which allows the quotient to be chosen acylindrically hyperbolic; we take G_0 to be such a quotient. Put $S = tJt^{-1}$. Since J is simple and nonabelian, J and S are perfect.

For $c \in J$ and $\ell \in S$, the commutator $[tct^{-1}, \ell]$ lies in $\mathfrak{D}_{G_0}(\Gamma)$ by (1), since t conjugates Γ into itself, c centralizes Γ, and $S \leq \Gamma$. As c ranges over J, the element tct^{-1} ranges over S, so $[S, S] \leq \mathfrak{D}_{G_0}(\Gamma)$, and S is perfect, so

$$S \leq \mathfrak{D}_{G_0}(\Gamma).$$

Proof of Theorem D. Let N be the normal closure of S in G_0; it is nontrivial because S is. By Lemma 7.2 applied to G_0, N, and $\Omega = \varnothing$, there is a surjective homomorphism $\varphi\colon G_0 \to Q$ with Q two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, and $\varphi(N) = Q$. The group Q is infinite because it is acylindrically hyperbolic, and it has property (T) as a quotient of G_0.

By (2), $\varphi(S) \leq \mathfrak{D}_Q(\varphi(\Gamma))$. The subgroup $\mathfrak{D}_Q(\varphi(\Gamma))$ is normal in Q, and the normal closure of $\varphi(S)$ is $\varphi(N) = Q$, so

$$\mathfrak{D}_Q(\varphi(\Gamma)) = Q.$$

Both Q and $\varphi(\Gamma)$ have property (T), as quotients of G_0 and Γ, respectively. By the last assertion of Theorem A, every homomorphism from Q to an MF group is trivial. If a quotient $\bar{Q}$ of Q is MF, then the quotient map $Q \to \bar{Q}$ is trivial, so $\bar{Q} = 1$. □

Corollary 7.3. *Let G be finitely presented, torsion-free, and acylindrically hyperbolic, and let $\Omega \subseteq G$ be finite. Then G has a two-generated, finitely presented, torsion-free, acylindrically hyperbolic quotient $\widetilde{G}$ with property* (T) *such that the quotient map is injective on Ω and every homomorphism from $\widetilde{G}$ to an MF group is trivial.*

Proof. Hull's common quotient theorem [Hu, Corollary 7.4], applied to G and to the group Q of Theorem D, gives a common quotient $\widetilde{G}$ that is acylindrically hyperbolic, with the map from G injective on a prescribed finite subset, since a torsion-free acylindrically hyperbolic group has no nontrivial finite normal subgroup. For finitely generated inputs, the proof of that corollary applies Theorem 7.1 to the free product $G * Q$ and then to the resulting quotient, each time with finitely many elements g_i, so $\widetilde{G}$ is torsion-free and finitely presented, as in the remark after Theorem 7.1. As a quotient of Q, the group $\widetilde{G}$ is two-generated and has property (T), and every homomorphism from $\widetilde{G}$ to an MF group pulls back to one from Q, so it is trivial. □

Corollary 7.4. *The algebra $C^*_{\mathrm{r}}(Q)$ is separable, unital, generated by two unitaries, simple, has a unique tracial state, has stable rank one, is stably finite, and is not MF.*

Proof. The group Q is countable, torsion-free, and acylindrically hyperbolic, so it contains a non-degenerate hyperbolically embedded subgroup [Os, Theorem 1.2] and has no nontrivial finite normal subgroup. Dahmani, Guirardel, and Osin give simplicity and uniqueness of the trace [DGO, Theorem 2.35], and Gerasimova and Osin give density of the invertible elements, which is stable rank one [GO, Theorem 1.1]. The algebra is separable and generated by the canonical unitaries of the two generators; its canonical trace is faithful, so it is stably finite; and it is not MF by Lemma 2.1, since Q is not MF (Theorem D). □

Questions

Corollary 2.5 makes every element of $\mathfrak{D}_G(L)$ invisible in normalized Hilbert–Schmidt norm. Theorem 2.7 upgrades that to triviality in a corona when the normal subgroup has property (T), and Theorem 5.1 when it acts on a finite-dimensional invariant subalgebra.

(1) Is the upgrade available for a locally finite normal subgroup inside $\mathfrak{D}_G(L)$, with no property-(T) hypothesis on it? Theorem 5.2 shows that local finiteness cannot simply be weakened to virtual abelianness.

(2) Theorem B assumes that the complementary idempotent $1 - st$ is full. Is $\mathrm{EL}_n(R)$ non-MF for every countable unital ring with $ts = 1 \neq st$? The first open case is the Jacobson algebra $\mathbb{F}_2\langle s, t \mid ts = 1\rangle$, where the ideal generated by $1 - st$ is spanned by the matrix units $s^i(1 - st)t^j$ and is therefore a union of finite subrings; the elementary group has the residually finite quotient $\mathrm{EL}_n(\mathbb{F}_2[z, z^{-1}])$, and its MF radical would be the finitary kernel of that symbol map. A positive answer to the first question settles this case, and every ring in which $1 - st$ has finite additive order.

(3) Is every amenable trace on a separable nuclear C^*-algebra quasidiagonal? The trace of Theorem E lives on a non-nuclear algebra, and for faithful traces on algebras satisfying the universal coefficient theorem the answer is positive [TWW].
(4) Is there a torsion-free group whose MF radical is nontrivial and proper? The exact radicals of Theorems E and F contain torsion, while the group of Theorem D has full MF radical.

Origin and authorship

In early August 2026 the author was using AI models on two questions raised by the nonsofic group of [OAI]: whether hyperlinearity implies soficity, and whether every group is hyperlinear. In one ChatGPT conversation, GPT-5.6 Sol, on "high" thinking mode, worked on weak MF (matricial field) approximations of a Clifford-twisted wreath product as a step toward hyperlinearity. Earlier in that conversation the author had asked whether to pivot to the MF question, and Sol had advised against it. The author gave Sol the existing progress on the hyperlinear problem and sent this prompt:

> "Achieve f{MF of the Clifford Kun–Thom extension }$\widetilde{W}$ in such a way that it archives our goals and ends everything, or, possibly, a different breakthrough, leading to achieving the final result. You got this!"

Sol answered with an MF counterexample: every norm-matrix model of the Clifford-twisted wreath product kills the Clifford sign. That was on August 9. Also on August 9, the models proved that the maximal group C^*-algebra of a strictly compressed Kazhdan group is infinite. Parallel chats, including one in "Pro" mode, and the agents working on the hyperlinear question, did not find the counterexample. By August 12 the models had reduced the obstruction to one commutator in a finitely presented group and had proved in Lean that it is not MF. They wrote the first draft of this paper that evening.

The author prompted the models, set the direction, retrieved the literature, and suggested the MF question. With the models, the author explored several routes to a non-MF group, shaped the current proof, and checked and revised the citations. The models drafted much of the text; the author wrote the rest and edited the whole for clarity and for the mathematical presentation. GPT-5.6 Sol in Codex, and Claude Fable 5 and Claude Opus 5 in Claude Code, wrote the Lean proofs. The whole exchange is in the repository's public commit history.[3]

The author consulted experts from August 13, and their comments shaped the revisions. Many models also reviewed drafts; Astra's reviews shortened the paper and simplified the proof of Theorem 2.7. We recommend Eckhardt's writeup [Ec]: a short, elementary proof that certain generalized wreath products are not MF.

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[3] https://github.com/SauersML/group-approximation

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